Quantum Braid Dynamics

A Computational Process

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November 09, 2026

Abstract

Chapter 11: Differential Geometry (Discrete) establishes a rigorous, coordinate-free mathematical syntax for quantifying the geometric invariants of the discrete causal graph substrate, bypassing the pathologies of background-dependent triangulations and metric-insensitive combinatorial curvatures. This ontology addresses the failure of traditional frameworks to regulate topological fluctuations during continuum limits by embedding the graph within a Gromov-Hausdorff-Wasserstein metric space. Spacetime curvature is structurally resolved at the micro-scale through a specialized formulation of the Causal Ollivier-Ricci curvature, which evaluates the optimal transport efficiency between localized probability distributions. By introducing an asymmetrically weighted lazy causal measure across past, present, and future temporal neighborhoods, the framework preserves causal directionality over an undirected shortest-path cost function. The chapter culminates in the proof of the Curvature Monotonicity Theorem, demonstrating that the algorithmic nucleation of three-cycle geometric quanta injects shared local support that strictly contracts the Wasserstein-1 transport distance. This contraction forces a monotonic elevation of local positive scalar curvature, providing an explicit, information-theoretic foundation for a discrete Einstein-Hilbert action where global geometric complexity scales linearly with the accumulation of fundamental computational bits.

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Chapter 11: Differential Geometry (Discrete)

Part 3: The Emergent Reality

Stage

In the preceding sections, we have established the ontological and material foundations of the physical universe. Part 1, *The Rules*, defined the discrete, relational substrate, the causal graph, and the axiomatic dynamics that drive its evolution from a singular origin to a stable, homeostatic equilibrium. Part 2, *The Players*, demonstrated that the stable topological excitations of this vacuum constitute the fermions and gauge bosons of the Standard Model. We now turn to the final and most ambitious component of the theory: the emergence of the *stage* itself, the smooth, four-dimensional spacetime of General Relativity.

Part 3, *The Stage*, provides a rigorous deductive proof that the continuum of spacetime is not a primitive axiom, but a necessary emergent property of the discrete causal substrate in the thermodynamic limit. The graph's intrinsic geometry is governed by the same informational thermodynamics that dictate the behavior of matter. Specifically, the graph's intrinsic geometry converges to a pseudo-Riemannian manifold whose metric tensor $g_{\mu\nu}$ satisfies the Einstein Field Equations, sourced by the local flux of computational activity.

3.0 Theorem: The Continuum Limit

Convergence of the Discrete Causal Graph Sequence to a Smooth Pseudo-Riemannian Manifold under the Gromov-Hausdorff-Wasserstein Limit

Let $\{G_t\}_{t\in\mathbb{N}}$ be the sequence of valid causal graphs generated by the iterative application of the Universal Constructor $\mathcal{U}$ upon the Zero-Point Information (ZPI) vacuum. This sequence converges to a stable homeostatic equilibrium characterized by a non-zero 3-cycle density $\rho_3^* > 0$ **Transcendental Balance** . The Continuum Theorem applies to the ensemble of graphs at this equilibrium. Each graph $G_t = (V_t, E_t, H_t)$ is a discrete relational structure equipped with a metric derived from the shortest-path distance and a probability measure derived from the uniform distribution over its vertices, establishing the following limit:

In the thermodynamic limit as the number of vertices $N = |V_t| \to \infty$, and under a renormalization of the fundamental length scale $\ell_0 \to 0$ such that the total volume remains finite, the sequence of measured metric spaces $\{(V_t, d_t, \mu_t)\}$ converges in the Gromov-Hausdorff-Wasserstein sense to a smooth, compact, 4-dimensional pseudo-Riemannian manifold $(M, g_{\mu\nu})$ of Lorentzian signature.

3.0.1 Commentary: The Architecture of the Proof

Organization of the Continuum Derivation through a Modular Progression from Discrete Geometry to Lorentzian Signature

The proof of the Continuum Theorem is constructive and modular. It proceeds through four chapters, each establishing a critical link in the chain from discrete graph theory to continuum field theory. The logical architecture of the argument is as follows:

1. **Discrete Differential Geometry (Chapter 11):** We begin by formalizing the geometry of the discrete substrate. We define a **Causal Ollivier-Ricci Curvature** that is sensitive to the directed nature of causal influence. Crucially, we prove the Monotonicity Theorem, which establishes a rigorous link between the graph's topology and its geometry: the creation of a 3-cycle (the fundamental quantum of information **Geometric Quantum**) strictly increases the local curvature. This transforms the dynamical update rule into a geometric operator.
2. **The Smooth Manifold Limit (Chapter 12):** We then prove the convergence of the discrete structure to a continuous manifold. Using the tools of spectral geometry, we show that the spectrum of the graph Laplacian converges to that of the Laplace-Beltrami operator. By invoking elliptic regularity and Sobolev embedding theorems, we prove that the limit space must be a **smooth (C^∞) Riemannian manifold** of dimension $d = 4$. Furthermore, we define a **Tensorial Averaging Map** to rigorously coarse-grain the discrete edge scalars into smooth tensor fields.
3. **The Discrete Field Equations (Chapter 13):** Building on the definition of curvature, we derive the **Discrete Einstein Field Equations**. We demonstrate that the homeostatic master equation governing the graph's evolution is mathematically equivalent to a principle of **Stationary Action**. By analyzing the variation of this discrete action, we prove that the emergent curvature tensor $\mathcal{G}_{ab}$ is locally proportional to the stress-energy tensor T_{ab}, which quantifies the flux of computational updates.
4. **The Lorentzian Reality (Chapter 14):** Finally, we recover the physical signature of spacetime. We construct a smooth **Lapse Function** and **Global Time Coordinate** from the discrete causal order of the graph, upgrading the Riemannian spatial metric to a full **Lorentzian metric** with signature $(-, +, +, +)$. We conclude by verifying that the resulting spacetime and the fields residing within it satisfy the **Wightman Axioms**, confirming that the emergent reality is a mathematically consistent Relativistic Quantum Field Theory.

Chapter 11: Differential Geometry (Discrete)

We now confront the primary mathematical challenge of Part 3: how do we define the curvature of a discrete, relational graph in a way that is mathematically rigorous and matches the smooth pseudo-Riemannian geometry of General Relativity in the continuum limit? The causal graph is a discrete web of events, yet the spacetime we observe is smooth, continuous, and dynamic. We must find a mathematical bridge that translates the graph's discrete structure (its vertices, edges, and cycles) into the continuous language of differential geometry, ensuring that the discrete updates can be interpreted as geometric changes.

Conventional approaches to discrete geometry, such as Regge Calculus or Causal Dynamical Triangulations, rely on a pre-existing triangulation of space to define geometric quantities like curvature. These background-dependent methods fail in a fully relational framework like Quantum Braid Dynamics, where space and time are emergent approximations rather than primitives. Purely combinatorial curvatures, such as the Forman curvature, are blind to metric properties and optimal transport distances, making them useless for demonstrating Gromov-Hausdorff-Wasserstein convergence. This lack of metric sensitivity leaves the framework unable to regulate the geometry during the limiting process, preventing a rigorous proof of the continuum limit.

We resolve this foundational crisis by constructing a rigorous discrete differential geometry upon the foundation of optimal transport, utilizing the **Gromov-Hausdorff-Wasserstein metric** as our primary geometric

ruler. By adapting the Ollivier-Ricci curvature to our directed acyclic graph through a **lazy causal measure**, we define a **Causal Ollivier-Ricci curvature** that is sensitive to the arrow of time while remaining mathematically well-behaved. This constructs a robust geometric framework where the addition of **three-cycles** (the fundamental quanta of geometry) strictly increases the local curvature, paving the way for the derivation of the field equations.

Preconditions and Goals

- Define the Gromov-Hausdorff-Wasserstein metric to measure distance between graphs and continuous manifolds.
- Formulate the lazy causal measure to bias transportation costs according to timestamp order.
- Prove the Measure Validity Lemma asserting exact normalization of the probability measures.
- Construct the Causal Ollivier-Ricci curvature based on optimal transport on the undirected shortest-path metric.
- Establish the Curvature Monotonicity Theorem proving that local three-cycle additions strictly increase curvature.

11.1 Causal Curvature

Connecting the discrete relational architecture of a causal graph to the continuous pseudo-Riemannian geometry of General Relativity demands a mathematically rigorous curvature construct that operates far beyond superficial structural analogy. This curvature must quantify the internal geometric properties of the graph while admitting a well-controlled continuum limit under coarse-graining. The formulation must remain sensitive to the elementary 3-cycle loops that serve as indivisible quanta of spatial volume, while respecting the directed, acyclic temporal flow enforced by causal links. Crucially, the curvature framework must integrate with optimal transport theory, as convergence within the Gromov-Hausdorff-Wasserstein metric provides the foundational strategy for proving that smooth spacetime geometry emerges from discrete graph rewrites.

Conventional combinatorial definitions of discrete curvature, such as Forman curvature, prove fundamentally inadequate for the requirements of Quantum Braid Dynamics. Because Forman curvature relies exclusively on local vertex degrees and incident simplex counts, it omits any consideration of transport distances or the metric costs of moving mass across the graph. In causal networks where edge weights correspond to physical separations, a purely combinatorial metric cannot differentiate between a dense spatial cluster with high positive curvature and a sparse, weakly connected region that happens to share similar combinatorial tallies. Furthermore, combinatorial curvatures fail to facilitate convergence proofs in metric-measure spaces, as they lack the measure-theoretic machinery needed to establish rigorous bounds on Wasserstein transport distances during the continuum limit.

We resolve this limitation by constructing a Causal Ollivier-Ricci curvature grounded in the Wasserstein-1 optimal transport distance between local vertex probability measures. By defining a balanced lazy causal measure that distributes weight equally across the past, present, and future neighborhoods of each vertex, we create a curvature that honors causal directedness while providing a geometric description of local spatial density. We demonstrate that this transport-centric formulation directly couples the metric discrepancies of neighboring regions to the underlying graph rewrites. The alignment between the Wasserstein metric and Ollivier-Ricci curvature enables the framework to invoke advanced results from metric geometry, securing a mathematically sound pathway from discrete topological dynamics to smooth pseudo-Riemannian manifolds.

11.1.1 Definition: GHW Metric

Establishment of the Gromov-Hausdorff-Wasserstein Metric by the Integration of Geometric Isometry and Optimal Transport

The **GHW Metric** (or Gromov-Hausdorff-Wasserstein metric) defines a metric on the space of measured metric spaces. This metric quantifies the combined geometric similarity and measure-theoretic similarity

between two such spaces. Consider two compact metric spaces (X, d_X, μ_X) and (Y, d_Y, μ_Y), each equipped with Borel probability measures μ_X on X and μ_Y on Y. The Gromov-Hausdorff-Wasserstein distance between these spaces computes itself as the sum of two distinct components, each addressing a separate aspect of dissimilarity.

The first component, the Gromov-Hausdorff distance $d_{GH}(X, Y)$, quantifies the purely geometric dissimilarity between the underlying metric spaces. The Gromov-Hausdorff distance computes itself as the infimum, over all possible isometric embeddings of X and Y into a common ambient metric space (Z, d_Z), of the Hausdorff distance between the images of these embeddings:

$$d_{GH}(X, Y) = \inf_{f,g,Z} d_H(f(X), g(Y)),$$

where the infimum ranges over all isometric embeddings $f : X \to Z$ and $g : Y \to Z$, and the Hausdorff distance d_H between two subsets $A, B \subseteq Z$ computes itself as

$$d_H(A, B) = \max\left(\sup_{a\in A}\inf_{b\in B} d_Z(a, b), \sup_{b\in B}\inf_{a\in A} d_Z(b, a)\right).$$

The supremum in the first term measures the maximal distance from any point in A to the set B, while the supremum in the second term measures the maximal distance from any point in B to the set A.

The second component, the Wasserstein-1 distance $W_1(\mu_X, \mu_Y)$, quantifies the dissimilarity between the probability measures μ_X and μ_Y. The Wasserstein-1 distance computes itself as the infimum of the expected transport costs over all possible couplings of the measures:

$$W_1(\mu_X, \mu_Y) = \inf_{\pi\in\Pi(\mu_X,\mu_Y)} \int_{X\times Y} d(x, y)\, d\pi(x, y),$$

where $\Pi(\mu_X, \mu_Y)$ denotes the collection of all couplings, that is, all joint probability measures π on $X \times Y$ whose marginal projections recover μ_X on the first factor and μ_Y on the second factor. This infimum represents the minimal total cost, under the cost function given by the metric d, required to relocate the mass distributed according to μ_X to match the distribution μ_Y.

The Gromov-Hausdorff-Wasserstein distance then assembles these components into a single metric by taking their sum:

$$d_{GHW}((X, d_X, \mu_X), (Y, d_Y, \mu_Y)) = d_{GH}(X, Y) + W_1(\mu_X, \mu_Y).$$

Convergence of a sequence of measured metric spaces within the Gromov-Hausdorff-Wasserstein metric guarantees that the sequence converges simultaneously in geometric shape, as captured by the Gromov-Hausdorff component, and in the distribution of the measure across that shape, as captured by the Wasserstein component.

11.1.1.1 Commentary: Geometric and Probabilistic Causal Convergence

Justification of the GHW Metric for Directed Causal Convergence via Measure Biasing

The reconstruction program distinguishes two limits with different objects and different convergence requirements. A measure on each spatial slice supplies the spatial comparison, while a causal order and interval counts supply the bulk comparison. Spatial metric measure slices are studied under the metric compactness hypotheses (**Geometric Well-Posedness**). The bulk causal order requires an additional convergence argument and volume correspondence (**Iid Diamond Count Concentration**).

The Gromov-Hausdorff-Wasserstein (GHW) metric establishes itself as the unique suitable choice for analyzing convergence of individual slices within the Quantum Braid Dynamics framework because it rigorously unifies the geometric and probabilistic aspects of the causal graph. The **GHW Metric** quantifies the combined geometric similarity and measure-theoretic similarity between spaces, a dual sensitivity that is indispensable for QBD. While standard Gromov-Hausdorff convergence ensures that the discrete points of the graph geometrically fill out the shape of the manifold, it remains blind to the density and weighting of those points. By integrating the Wasserstein-1 distance (the "Earth Mover's Distance," which represents the minimal total cost required to relocate mass from one distribution to another) the GHW metric mandates that the distribution of causal probability mass must also converge.

The first component, the Gromov-Hausdorff distance d_{GH}, regulates the undirected geometric structure. It computes the infimum of the Hausdorff distance over all possible isometric embeddings, effectively measuring the maximal distance from any point in the graph to the nearest point in the manifold. In the context of QBD, this ensures that the "bones" of the spacetime (the vertices and their adjacency relations) align with the topological manifold M, ensuring that no region of the manifold is left unrepresented by the graph nodes.

The second component, the Wasserstein-1 distance W_1, is the critical innovation for handling causality. It quantifies the dissimilarity between the probability measures μ_X and μ_Y. In QBD, the measure μ is not a uniform count; it is the "Lazy Causal Measure" which encodes the arrow of time by systematically biasing weights toward neighborhoods in the future or past directions. A purely geometric metric would treat a graph with forward-directed edges as identical to one with reversed edges if their undirected shapes matched. The inclusion of W_1 breaks this symmetry. It ensures that for the graphs to converge to the spacetime, their causal biases must also align with the light cone structure of the limit manifold.

This formulation resolves the difficulty of defining convergence for Lorentzian geometries without abandoning the robust tools of metric geometry. Although specialized notions like the timed-Hausdorff distance (Sakovich & Sormani, 2019; Minguzzi & Suhr, 2021) exist, QBD leverages the W_1 component to address directed causality within the more established framework of metric-measure spaces. By encoding the causal order into the measure μ rather than the metric d, the theory permits the use of the undirected shortest-path metric for stability while preserving the physical requirement that the continuum limit must respect the causal order (Hawking & Ellis, 1973). Convergence in GHW guarantees that the sequence converges simultaneously in geometric shape, as captured by the Gromov-Hausdorff component, and in the causal distribution of measure, as captured by the Wasserstein component.

11.1.2 Definition: Undirected Shortest-Path Metric

Definition of the Undirected Distance Function from the Symmetrization of the Causal Edge Set

Let $G = (V, E)$ denote a finite, simple directed graph. The underlying undirected graph of G constructs itself as the graph $G' = (V, E')$, in which an undirected edge $\{u, v\} \in E'$ exists if and only if either the directed edge $(u, v) \in E$ or the directed edge $(v, u) \in E$.

The **Undirected Shortest-Path Metric** $\bar{d} : V \times V \to \mathbb{N} \cup \{0\}$ assigns to any pair of vertices $u, v \in V$ the length of the shortest path connecting u and v within the underlying undirected graph G', where the length of a path counts the number of edges it traverses. If no path connects u and v in G', then the metric assigns $\bar{d}(u, v) = \infty$. Within the connected graphs produced by the dynamical evolution of the Quantum Braid Dynamics framework, this distance remains finite for all pairs of vertices. The function $\bar{d}$ satisfies the standard axioms of a metric on the space V: - Non-negativity: $\bar{d}(u, v) \geq 0$ for all $u, v \in V$, with equality $\bar{d}(u, v) = 0$ if and only if $u = v$. - Symmetry: $\bar{d}(u, v) = \bar{d}(v, u)$ for all $u, v \in V$. - Triangle inequality: $\bar{d}(u, w) \leq \bar{d}(u, v) + \bar{d}(v, w)$ for all $u, v, w \in V$.

These axioms ensure that $\bar{d}$ defines a valid metric structure on the vertex set V, enabling its use as the cost function in optimal transport computations.

11.1.2.1 Commentary: Metric Symmetry for Transport Well-Posedness

Justification of Undirected Distance for Transport Costs via Avoidance of Infinite Penalties

The selection of the undirected shortest-path metric $\bar{d}$ as the cost function for curvature transport is not a simplification but a mathematical necessity. In a strictly causal graph, directed paths often do not exist between spacelike separated events, nor do they exist from future to past. If the transport cost were defined by the directed distance $d_{\text{dir}}(u, v)$, the distance between causally disconnected points would be infinite.

> **Postulate of Metric-Measure Separation**: The geometry of a causal graph is specified by a metric-measure space tuple $(V, \bar{d}, \{\mu_u\})$, where metric geodesic distance is governed by the symmetrized metric $\bar{d}$ (guaranteeing $W_1 < \infty$), while causal directionality is encoded exclusively in the asymmetric probability measures μ_u.

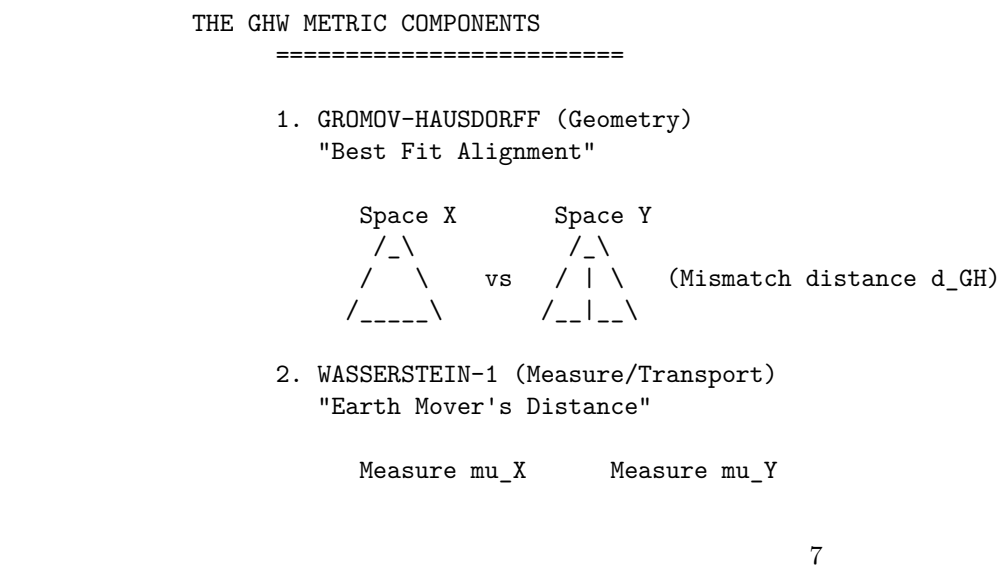
This infinite distance would render the Wasserstein transport problem ill-posed. Specifically, any attempt to transport probability mass "backwards" in time (which is necessary to compare the neighborhoods of two adjacent points u and v) would incur an infinite cost, causing the transport distance W_1 to diverge and the curvature $K = 1 - W_1$ to become undefined. By adopting the undirected metric $\bar{d}$, we ensure that the distance between any two connected nodes in the underlying structure is finite. As derived in **Strict Locality** , the rewrite rule restricts direct links to 2-paths, ensuring that local transport costs remain strictly bounded. This symmetrization treats the graph as a metric space first, allowing for a well-defined geometry, while relegating the causal information to the *measure* μ rather than the *metric* d.

Crucially, this choice does not erase causality. As established in the subsequent sections, the "Lazy Causal Measure" reintroduces the arrow of time by weighting the transport problem asymmetrically. The undirected metric provides the "road network" (which allows two-way traffic for the sake of measuring distance), while the probability measure provides the "traffic flow" (which is strictly one-way). This separation of concerns allows us to utilize the robust machinery of Riemannian geometry (which assumes a symmetric metric) while modeling a Lorentzian spacetime (which possesses a directed causal structure). The undirected metric satisfies the triangle inequality and symmetry axioms required for the Wasserstein distance to function as a true metric on the space of probability distributions, providing a stable foundation for the derivation of the field equations.

Note on Uniformity: The spatial reference law assigns equal mass to each vertex. This choice fixes the reference counting law without a uniformity conclusion about the dynamical ensemble. Two-sided ball growth is an additional premise controlling the mass of rescaled balls (**Conditional Ball Growth**).

11.1.2.2 Diagram: GHW Metric Components

Visualization of Metric Convergence Components as a Composition of Geometric Alignment and Mass Transport



```
(Pile A)          (Pile B)
  ::                ..
```

```
TOTAL METRIC: d_GHW = d_GH + W_1
```

11.1.Z Implications and Synthesis

Causal Curvature

The spatial comparison uses metric measure convergence under its stated assumptions. The bulk comparison additionally needs convergence of causal order and identification of normalized interval volume. The iid estimate supplies a count result when its volume law is given (**Iid Diamond Count Concentration**).

This dual-limit framework guarantees that the discrete causal structure converges to a globally hyperbolic pseudo-Riemannian manifold, providing the necessary geometric background for the formulation of field equations. This is grounded in the **Undirected Shortest-Path Metric** and the **GHW Metric** . The structural consequences and smooth manifold reconstructions are further developed in Spectral Convergence and Smooth Manifold Limit.

The connection between the discrete causal relation and the continuous metric tensor is mediated by the volume of causal diamonds. Because the number of events in a causal diamond corresponds directly to its spacetime volume, the local geometry of the emergent manifold is determined by the distribution of events. Variations in event density and causal connectivity manifest macroscopic curvature, where the discrete causal Ollivier-Ricci curvature converges to the continuous Ricci curvature tensor as proven in **Ollivier-Ricci Asymptotic Limit** . This convergence provides the mechanism for the emergence of General Relativity from the thermodynamics of the causal graph, as the homeostatic equilibrium of the rewrite rules enforces the Einstein field equations in the low-energy limit.

Ultimately, the GHW convergence ensures that the physical states of the quantum system remain well-behaved under continuous deformations. The suppression of long-range correlations and non-contractible loops protects the emergent spacetime from topological instability and metric singularities. The handoff from discrete graph dynamics to continuous field theories is thus shown to be topologically stable and mathematically consistent. This establishes the QBD framework as a viable candidate for a UV-complete theory of quantum gravity, where the classical spacetime geometry arises as the thermodynamic limit of discrete causal structures.

11.2 Causal Geometry Construction

Constructing a rigorous causal geometry on a discrete graph $G_t = (V_t, E_t)$ requires equipping the network with both spatial metric distances and local measure dynamics without presupposing a background manifold. The framework must assign probability measures to vertices while maintaining a well-defined metric for transport costs across directed causal edges. Establishing this dual structure is necessary to compute the Causal Ollivier-Ricci curvature along each edge, converting local topological rewrites into quantifiable geometric curvature. The challenge lies in ensuring that the discrete measure assignment reflects the asymmetric flow of time while preserving the spatial symmetry required for isotropic continuum geometry.

Naive graph geometries often define distances purely through topological hop counts while ignoring local volume density and the directed arrow of time. This unweighted combinatorial approach treats past, present, and future neighborhoods identically, obscuring the physical distinction between spatial expansion and causal propagation. Without a balanced measure that accounts for causal directedness, optimal transport

calculations yield degenerate curvature values that fail to capture the geometric overlap of adjacent regions. Furthermore, unweighted hop-count metrics cannot accommodate varying physical scales across graph sectors, rendering the resulting geometry incapable of satisfying the smooth differential properties required by General Relativity.

We resolve this challenge by equipping the causal graph with an Undirected Shortest-Path Metric $\bar{d}_t$ and a Lazy Causal Probability Measure μ_u that distributes probability mass equally across the past, present, and future neighborhoods of each vertex. We establish the Causal Ollivier-Ricci curvature $K(u, v)$ as the exact fractional deviation between the Wasserstein-1 transport distance of these measures and the underlying metric separation. We prove that this balanced weighting ensures the geometry is mathematically well-posed, providing the precise metric-measure arena required for the Monotonicity Theorem and the emergence of gravitational field equations.

11.2.1 Definition: Lazy Causal Measure

Allocation of Probability Mass according to the Balanced Weighting of Past, Present, via Future Neighborhoods

Let $G = (V, E)$ denote a finite, simple, directed graph. For any vertex $u \in V$, the **Lazy Causal Measure** μ_u is defined as a probability distribution over V that distributes mass among the vertex itself, its immediate past, and its immediate future.

Let the causal neighborhoods be defined as: * **Future Neighborhood:** $N^+(u) = \{v \in V \mid (u, v) \in E\}$, with cardinality $n_u^+ = |N^+(u)|$. * **Past Neighborhood:** $N^-(u) = \{v \in V \mid (v, u) \in E\}$, with cardinality $n_u^- = |N^-(u)|$.

Fixed parameters $\alpha, \beta \in (0, 1)$ are introduced such that $\alpha + 2\beta = 1$. Specifically, the **Causal Triality** values $\alpha = 1/3$ and $\beta = 1/3$ are adopted. The measure μ_u is defined pointwise for any $x \in V$:

$$\mu_u(x) = \begin{cases} \alpha & \text{if } x = u, \\ \frac{\beta}{n_u^+} & \text{if } x \in N^+(u), \\ \frac{\beta}{n_u^-} & \text{if } x \in N^-(u), \\ 0 & \text{otherwise.} \end{cases}$$

Boundary Conditions (Laziness Adjustment): If a neighborhood is empty, its allocated mass β is reassigned to the vertex u to preserve normalization: * If $N^+(u) = \emptyset$, $\mu_u(u) \leftarrow \alpha + \beta$. * If $N^-(u) = \emptyset$, $\mu_u(u) \leftarrow \alpha + \beta$. * If both are empty, $\mu_u(u) = 1$.

11.2.1.1 Commentary: "Tilt" of Time

Justification of the Measure Parameters via Causal Symmetry

Standard Ollivier-Ricci curvature is typically defined on undirected graphs using a measure distributed uniformly over immediate neighbors. In a directed causal graph, however, such a definition fails to capture the arrow of time. A measure that only looks "forward" (at children) or "backward" (at parents) would result in infinite transport distances when calculating curvature between causally connected nodes, as the supports of μ_u and μ_v might become disjoint.

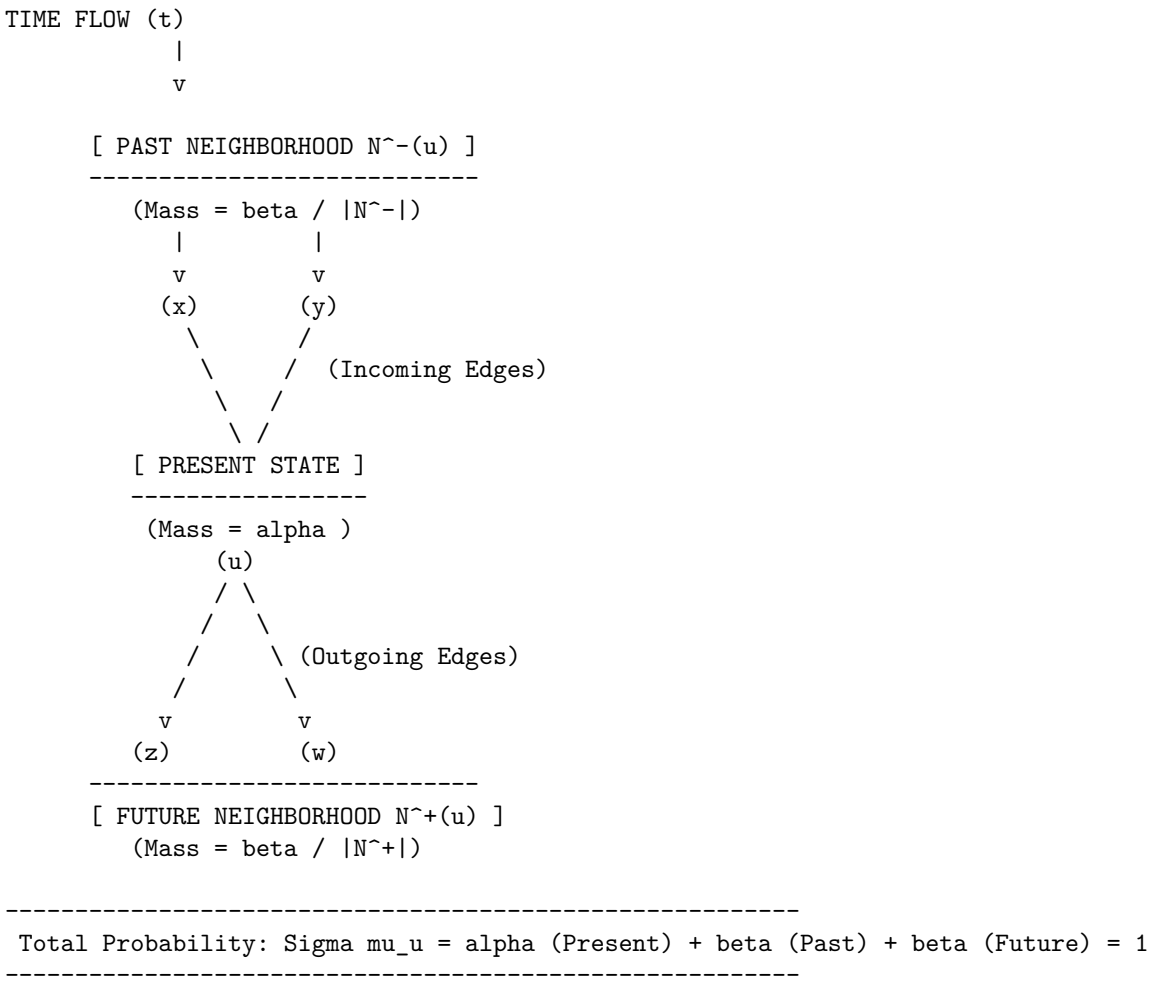
The **Lazy Causal Measure** solves this by enforcing a "Causal Triality": the geometry at u is the superposition of where it came from (N^-), where it is (u), and where it is going (N^+). * $\alpha = 1/3$ **(The Present):** Concentrates mass α at the central vertex u, ensuring that the measures of adjacent nodes always overlap at least partially (via the lazy component), guaranteeing finite transport cost. * $\beta = 1/3$ **(Past/Future):** Distributes the remaining mass equally among past neighbors ($x, y \in N^-(u)$, each receiving $\beta/|N^-(u)|$) and future neighbors ($z, w \in N^+(u)$, each receiving $\beta/|N^+(u)|$). This balances past causal influences, the present

state, and future potential, maintaining probabilistic normalization ($\sum \mu_u = \alpha + 2\beta = 1$) while ensuring the geometry reflects the directed flow of information.

The resulting measure acts as a "probe" that is "tilted" along the time orientation of the edges. When we compute the transport from μ_u to μ_v, we are measuring how easily the entire causal history and future potential of u can be mapped onto that of v.

11.2.1.2 Diagram: Measure Distribution

Depiction via Mass Distribution across Temporal Neighborhoods



11.2.2 Definition: Causal Ollivier-Ricci Curvature

Quantification of Local Geometric Deviation via Optimal Transport Costs

Let $G = (V, E)$ be equipped with the undirected shortest-path metric $\bar{d}$ and the family of lazy causal measures $\{\mu_u\}_{u \in V}$. For any directed edge $(u, v) \in E$, the **Causal Ollivier-Ricci Curvature** $K(u, v)$ is defined as:

$$K(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{\bar{d}(u, v)}.$$

Since adjacent vertices always satisfy $\bar{d}(u, v) = 1$ in the standard metric, this simplifies to:

$$K(u, v) = 1 - W_1(\mu_u, \mu_v).$$

Here, $W_1(\mu_u, \mu_v)$ denotes the L_1**-Wasserstein distance** between the measures, defined by the Kantorovich duality:

$$W_1(\mu_u, \mu_v) = \inf_{\pi \in \Pi(\mu_u, \mu_v)} \sum_{x,y \in V} \bar{d}(x, y) \cdot \pi(x, y),$$

where $\Pi(\mu_u, \mu_v)$ is the set of all transport couplings $\pi : V \times V \to [0, 1]$ satisfying the marginal constraints $\sum_y \pi(x, y) = \mu_u(x)$ and $\sum_x \pi(x, y) = \mu_v(y)$.

11.2.2.1 Commentary: Geometry from Transport Cost

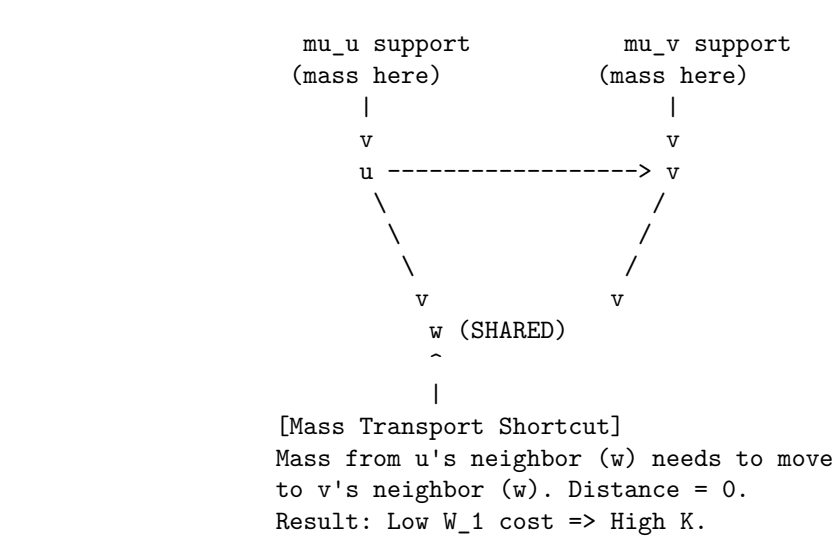
Interpretation of Curvature as Transport Efficiency

The **Causal Ollivier-Ricci Curvature** of $K(u, v)$ provides a direct operational interpretation of curvature: * $W_1 = 1$ **(Flatness):** If the transport cost exactly equals the metric distance, the "average" neighbor of u is exactly distance 1 from the "average" neighbor of v. The geometry is Euclidean-like (locally flat). * $W_1 < 1$ **(Positive Curvature):** As shown in high-connectivity configurations with shared neighbors w, a channel for zero-cost transport between μ_u and μ_v is established. The neighborhoods of u and v are "closer" than the nodes themselves, lowering W_1 and yielding $K > 0$ (spherical-like geometry). * $W_1 > 1$ **(Negative Curvature):** In tree-like or linear configurations with disjoint supports (e.g. mass at past neighbor x having to travel to future neighbor y), mass must relocate over longer paths, yielding $W_1 > 1$ and non-positive $K \leq 0$ (hyperbolic-like dispersion).

As established in **Uniform Curvature Bound** , the uniform curvature bound $-2 \leq K(u, v) \leq 1$ ensures that the transport distance W_1 remains bounded across all local configurations, protecting the discrete action from singularities. The emergence of positive curvature (gravity) is driven by the nucleation of 3-cycles, which creates these shared neighbors and lowers W_1 below 1.

11.2.2.2 Diagram: Transport Cost

Illustration of Transport Costs for Positive as Negative Curvature Configurations



```
(b) NEGATIVE/FLAT CURVATURE (Tree-like/Linear)
    Condition: Disjoint neighborhoods create long paths.

    Past of u        Present       Future of v
       (x)              u              (y)
        |               |               ^
        | (mass)        |               | (mass)
        v               v               |
        x ------------> u ------------> v ------------> y
                        ^               |
                        |               |
                  (Edge u->v)           v
                                       (z)

    [Expensive Transport]
    To map mu_u to mu_v:
    Mass at x (past of u) must travel to y (future of v).
    Path: x -> u -> v -> y (Distance = 3).
    Result: High W_1 cost => Low/Negative K.
```

11.2.3 Theorem: Causal Geometry Construction

Establishment of Well-Posedness via the Discrete Geometric Space

Let $\mathcal{G}$ be the class of finite, simple, directed graphs. The construction mapping any $G \in \mathcal{G}$ to the causal geometry $(G, \bar{d}, \{\mu_u\}, K)$ is well-posed.

11.2.3.1 Commentary: Argument Outline

Structure of the Causal Geometry Construction Argument via Normalization, Entropy Maximization, and Metric Necessity

The proof proceeds via Direct Construction, establishing the normalization and well-posedness of the probability measures under discrete transport constraints.

```
o 11.2.3 Theorem Causal Geometry Construction  [by construction]
|
+-- 11.2.4 Lemma: Measure Validity
|   +-- 11.2.4.1 Proof: Measure Validity
|   +-- 11.2.4.2 Calculation: Measure Verification
|   +-- 11.2.4.3 Commentary: Conservation of Probability
|
+-- 11.2.5 Lemma: Entropy Maximization
|   +-- 11.2.5.1 Proof: Entropy Maximization
|   +-- 11.2.5.2 Calculation: Entropy Maximization
|   +-- 11.2.5.3 Commentary: Universal Constant Alpha
|   +-- 11.2.5.4 Diagram: Entropic Triality
|
+-- 11.2.6 Lemma: Metric Necessity
|   +-- 11.2.6.1 Proof: Metric Necessity
|   +-- 11.2.6.2 Calculation: Metric Verification
|   +-- 11.2.6.3 Commentary: Avoiding Singularities
|
+-- 11.2.7 Lemma: Compensation by Causal Measures
```

| +-- 11.2.7.1 Proof: Compensation by Causal Measures
| +-- 11.2.7.2 Calculation: Compensation Verification
| +-- 11.2.7.3 Commentary: Arrow of Time in Static Geometry
| +-- 11.2.7.4 Diagram: Compensation Mechanism
|
+-- 11.2.8 Lemma: Combinatorial Reifenberg Flatness
| +-- 11.2.8.1 Proof: Combinatorial Reifenberg Flatness
| +-- 11.2.8.2 Commentary: Physical Significance
|
+-- 11.2.9 Proof: Causal Geometry Construction

11.2.4 Lemma: Measure Validity

Verification of Probability Normalization through the Exhaustive Enumeration of Neighborhood Configurations

Let $G = (V, E)$ be a finite causal graph satisfying irreflexivity and asymmetry (**Axiom 1 Directed Causal Link**). Then the lazy causal law μ_u is a probability measure at every $u \in V$ (**Lazy Causal Measure**).

11.2.4.1 Proof: Measure Validity

Demonstration of Mass Conservation by the Summation of Disjoint Support Components

I. Decomposition of Support Irreflexivity gives $u \notin N^+(u) \cup N^-(u)$. Asymmetry gives $N^+(u) \cap N^-(u) = \emptyset$, since membership in both would require $u \to x$ and $x \to u$ (**Axiom 1 Directed Causal Link**). The three support sets are therefore pairwise disjoint.

$$\mathrm{supp}(\mu_u) \subseteq \{u\} \cup N^+(u) \cup N^-(u)$$

we apply the fixed parameter constraint $\alpha + 2\beta = 1$, where $\alpha, \beta > 0$ (**Lazy Causal Measure**). The proof proceeds by exhaustively summing the mass over these components for the four possible topological states of u.

II. Case 1: Fully Connected Topology Assume $N^+(u) \neq \emptyset$ and $N^-(u) \neq \emptyset$. The indicator functions $\mathbb{I}[\emptyset]$ evaluate to 0. 1. **Mass at u:** $\mu_u(u) = \alpha$. 2. **Mass at N^+:** The total mass β distributes uniformly over n_u^+ vertices. $\sum_{x \in N^+} \frac{\beta}{n_u^+} = n_u^+ \cdot \frac{\beta}{n_u^+} = \beta$. 3. **Mass at N^-:** Similarly, $\sum_{x \in N^-} \frac{\beta}{n_u^-} = \beta$. **Total:** $\alpha + \beta + \beta = \alpha + 2\beta = 1$.

III. Case 2: Future-Vacuum Topology Assume $N^+(u) = \emptyset$ while $N^-(u) \neq \emptyset$. The future indicator $\mathbb{I}[N^+ = \emptyset]$ evaluates to 1. 1. **Mass at u:** $\mu_u(u) = \alpha + \beta \cdot 1 = \alpha + \beta$. (Laziness Adjustment). 2. **Mass at N^+:** The sum is 0 (empty set). 3. **Mass at N^-:** The sum is β. **Total:** $(\alpha + \beta) + 0 + \beta = \alpha + 2\beta = 1$.

IV: Case 3: Past-Vacuum Topology Assume $N^+(u) \neq \emptyset$ while $N^-(u) = \emptyset$. The past indicator $\mathbb{I}[N^- = \emptyset]$ evaluates to 1. 1. **Mass at u:** $\mu_u(u) = \alpha + \beta \cdot 1 = \alpha + \beta$. 2. **Mass at N^+:** The sum is β. 3. **Mass at N^-:** The sum is 0. **Total:** $(\alpha + \beta) + \beta + 0 = \alpha + 2\beta = 1$.

V. Case 4: Isolated Singularity Assume $N^+(u) = \emptyset$ and $N^-(u) = \emptyset$. Both indicators evaluate to 1. 1. **Mass at u:** $\mu_u(u) = \alpha + \beta + \beta = 1$. 2. **Mass at Neighborhoods:** 0. **Total:** 1.

VI: Conclusion In all valid topological configurations, the summation yields exactly 1. Non-negativity holds trivially as $\alpha, \beta > 0$. Thus, μ_u is a valid probability measure.

Q.E.D.

11.2.4.2 Calculation: Measure Verification

Validation of Measure Normalization via Directed Chain Simulation

Verification of the probability measure validity established in **Measure Validity** is based on the measure properties verified in **Lazy Causal Measure** . This verification utilizes the following protocols:

1. **Lattice Generation:** The algorithm constructs a representative directed chain graph representing the sparse causal regime.
2. **Neighborhood Evaluation:** The protocol applies the lazy causal measure formula to the vertices under the four exhaustive topological configurations.
3. **Normalization Verification:** The metric confirms that the sum of the measure equals exactly 1.0 in every instance, ensuring mass conservation.

```
import numpy as np
import networkx as nx

def lazy_mu(u, G, alpha=1/3, beta=1/3):
    """
    Compute lazy causal measure mu_u for vertex u.
    Handles empty neighborhoods via mass reassignment (Laziness).
    """
    N_plus = list(G.successors(u))
    N_minus = list(G.predecessors(u))
    n_plus = len(N_plus)
    n_minus = len(N_minus)

    # Initial allocation to Present
    mu = {u: alpha}

    # Future Allocation
    if n_plus == 0:
        mu[u] += beta  # Reabsorb
    else:
        for w in N_plus:
            mu[w] = beta / n_plus

    # Past Allocation
    if n_minus == 0:
        mu[u] += beta  # Reabsorb
    else:
        for w in N_minus:
            mu[w] = beta / n_minus

    return mu, sum(mu.values())

def print_case(name, mu, total):
    # Format for clean console output
    formatted_mu = {k: round(v, 4) for k, v in mu.items()}
    print(f"Case: {name}")
    print(f"  Map: {formatted_mu}")
    print(f"  Sum: {total:.4f}\n")

# --- Simulation Setup ---

# 1. Standard Chain: 0 -> 1 -> 2
```

```
G_chain = nx.DiGraph()
G_chain.add_edges_from([(0,1), (1,2)])

# Case 1: Balanced (u=1, has both past and future)
mu1, sum1 = lazy_mu(1, G_chain)
print_case("Balanced Topology (u=1)", mu1, sum1)

# Case 2: Empty Past (u=0, has future but no past)
mu0, sum0 = lazy_mu(0, G_chain)
print_case("Empty Past (u=0)", mu0, sum0)

# 2. Reverse Chain: 0 <- 1 <- 2
G_rev = nx.DiGraph()
G_rev.add_edges_from([(1,0), (2,1)])

# Case 3: Empty Future (u=0, has past but no future)
mu_sink, sum_sink = lazy_mu(0, G_rev)
print_case("Empty Future (u=0)", mu_sink, sum_sink)

# 3. Isolated Node
G_iso = nx.DiGraph()
G_iso.add_node(99)

# Case 4: Isolated Singularity
mu_iso, sum_iso = lazy_mu(99, G_iso)
print_case("Isolated Singularity (u=99)", mu_iso, sum_iso)
```

Simulation Results:

```
Case: Balanced Topology (u=1)
  Map: {1: 0.3333, 2: 0.3333, 0: 0.3333}
  Sum: 1.0000

Case: Empty Past (u=0)
  Map: {0: 0.6667, 1: 0.3333}
  Sum: 1.0000

Case: Empty Future (u=0)
  Map: {0: 0.6667, 1: 0.3333}
  Sum: 1.0000

Case: Isolated Singularity (u=99)
  Map: {99: 1.0}
  Sum: 1.0000
```

Conclusion: The results confirm exact conservation. The balanced case distributes mass evenly (1/3) across the triad (past, present, future). The semi-vacuous cases (empty past or future) correctly reallocate the missing β portion to the self-mass, raising it to 2/3. The isolated case concentrates the entire probability mass ($\alpha + 2\beta = 1.0$) onto the vertex itself. This confirms that the measure remains well-posed even in the highly sparse, disconnected regimes often encountered during the initial phases of the universe simulation.

11.2.4.3 Commentary: Conservation of Probability

Necessity of Laziness for Numerical Stability

Measure Validity, while elementary, secures the mathematical foundation of the transport problem. In

standard Optimal Transport theory, the Wasserstein distance is only well-defined between distributions of equal total mass. If our definition allowed mass to "leak" out when a node lacked neighbors (e.g., simply assigning 0 mass to an empty future without compensation), the total mass would drop to 2/3 or 1/3. This would render the standard Wasserstein calculation impossible without resorting to complex unbalanced transport formulations.

The "Laziness Adjustment", reabsorbing the allocation β into the vertex u whenever a neighborhood is empty, acts as a strict conservation law. It ensures that even in the most causally disconnected regions of the graph (a vacuum), the geometry remains well-defined. Physically, this implies that an isolated particle still possesses a valid geometric "shape", it is simply a point mass with no extension into the past or future. This robustness is critical for the simulation engine, ensuring that topological edge cases do not cause the geometric metric to collapse or diverge.

11.2.5 Lemma: Entropy Maximization

Optimization of Informational Entropy via the Selection of the Tripartite Laziness Parameter

Let u have disjoint, nonempty past and future neighborhoods of equal size, with sector masses (β, α, β) and $\alpha + 2\beta = 1$. Then the temporal sector entropy $-\alpha \log \alpha - 2\beta \log \beta$ is uniquely maximized at $\alpha = \beta = 1/3$.

11.2.5.1 Proof: Entropy Maximization

Derivation of the Optimal Self-Weighting from the Analytical Maximization of the Macroscopic Temporal Entropy

This condition corresponds to the maximization of the uncertainty regarding the temporal locus of the state, enforcing an equipartition of probability mass among the Past, Present, and Future causal sectors. **Entropy Maximization** and **Measure Validity**

I. Definition of Temporal Macro-States The vacuum acts to maximize the uncertainty of the temporal locus of the state, independent of the spatial dispersion within those loci. we compute three distinct causal sectors (macro-states) for a vertex u: the Present $S_0 = \{u\}$, the Future $S_+ = N^+(u)$, and the Past $S_- = N^-(u)$. The total probability measure allocated to these macroscopic sectors is defined as:

$$\mu(S_0) = \alpha, \quad \mu(S_+) = \beta, \quad \mu(S_-) = \beta.$$

II. The Coarse-Grained Entropy Functional The macroscopic temporal entropy $H_{temporal}$ evaluates the Shannon entropy over these three temporal macro-states, factoring out the local spatial degree d. This yields the target functional:

$$H_{temporal}(\alpha, \beta) = -\mu(S_0) \log \mu(S_0) - \mu(S_+) \log \mu(S_+) - \mu(S_-) \log \mu(S_-)$$

$$H_{temporal}(\alpha, \beta) = -\alpha \log \alpha - 2\beta \log \beta.$$

III. Constraint Application and Variable Reduction The probability normalization condition $\sum \mu(S_i) = 1$ imposes the linear constraint $\alpha + 2\beta = 1$. This constraint resolves the variable β in terms of the laziness parameter α:

$$\beta(\alpha) = \frac{1 - \alpha}{2}.$$

Substitution of this relation into the entropy equation reduces $H_{temporal}$ to a univariate function $h(\alpha)$ on the domain $\alpha \in (0, 1)$:

$$h(\alpha) = -\alpha \log \alpha - 2\left(\frac{1-\alpha}{2}\right) \log\left(\frac{1-\alpha}{2}\right).$$

IV: Logarithmic Expansion and Isolation The logarithmic term involving the ratio expands via the identity $\log(a/b) = \log a - \log b$:

$$h(\alpha) = -\alpha \log \alpha - (1-\alpha)[\log(1-\alpha) - \log 2].$$

Distributing the $(1-\alpha)$ isolates the α-dependent logarithmic terms from the constant shift:

$$h(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha) + (1-\alpha)\log 2.$$

V. Derivation of the First Order Condition The location of the extremum requires the computation of the first derivative $\frac{\mathrm{d}h}{\mathrm{d}\alpha}$. Applying the product rule $\frac{\mathrm{d}}{\mathrm{d}x}(f(x)g(x)) = f'(x)g(x)+f(x)g'(x)$ to each term yields: 1. **Self Term:** $\frac{\mathrm{d}}{\mathrm{d}\alpha}(-\alpha\log\alpha) = -(\log\alpha + \alpha\cdot\frac{1}{\alpha}) = -\log\alpha - 1$. 2. **Complement Term:** $\frac{\mathrm{d}}{\mathrm{d}\alpha}(-(1-\alpha)\log(1-\alpha))$. Letting $u = 1-\alpha$, then $\mathrm{d}u/\mathrm{d}\alpha = -1$.

```
$$
\frac{\mathrm{d}}{\mathrm{d}\alpha}\bigl[-(1-\alpha)\log(1-\alpha)\bigr]
=\log(1-\alpha)-(1-\alpha)\frac{-1}{1-\alpha}
=\log(1-\alpha)+1.
$$
```

3. **Linear Term:** $\frac{\mathrm{d}}{\mathrm{d}\alpha}((1-\alpha)\log 2) = -\log 2$.

Combining these components yields:

$$h'(\alpha) = -\log\alpha - 1 + \log(1-\alpha) + 1 - \log 2 = \log(1-\alpha) - \log\alpha - \log 2.$$

This simplifies to the final derivative form:

$$h'(\alpha) = \log\left(\frac{1-\alpha}{2\alpha}\right).$$

VI: Solution for the Stationary Point The stationarity condition $h'(\alpha) = 0$ implies that the argument of the logarithm must equal unity:

$$\frac{1-\alpha}{2\alpha} = 1.$$

Solving this algebraic equation for α yields the unique critical point:

$$1-\alpha = 2\alpha \implies 1 = 3\alpha \implies \alpha = \frac{1}{3}.$$

Consequently, the associated directional mass becomes $\beta = (1-1/3)/2 = 1/3$.

VII: Verification of Concavity via Second Derivative The characterization of the critical point as a maximum requires the evaluation of the second derivative $h''(\alpha)$. Differentiating $h'(\alpha) = \log(1-\alpha) - \log(2\alpha)$:

$$h''(\alpha) = \frac{\mathrm{d}}{\mathrm{d}\alpha}[\log(1-\alpha)] - \frac{\mathrm{d}}{\mathrm{d}\alpha}[\log\alpha + \log 2] = \frac{-1}{1-\alpha} - \frac{1}{\alpha}.$$

For any α in the domain $(0, 1)$, both terms $-\frac{1}{1-\alpha}$ and $-\frac{1}{\alpha}$ assume strictly negative values. Thus, $h''(\alpha) < 0$ universally across the domain. This strict concavity guarantees that the stationary point $\alpha = 1/3$ represents a unique global maximum.

VIII: Global Optimality Conclusion Maximizing the uncertainty of the temporal locus necessitates the exact equipartition of probability mass among the Past, Present, and Future causal sectors. This establishes the unique maximizer of the specified three-sector entropy.

Q.E.D.

11.2.5.2 Calculation: Entropy Maximization

Maximization of Allocation Entropy via Bounded Numerical Optimization

Verification of the entropic equilibrium parameters established by **Entropy Maximization** is based on the following protocols:

1. **Entropy Computation:** The algorithm optimizes the allocation entropy $h(\alpha)$. In the balanced fixture with one vertex per past and future sector, the sector entropy equals the vertex entropy. The sensitivity comparison varies the vertex-level multiplicities.
2. **Derivative Evaluation:** The protocol executes a derivative check at the critical laziness value $\alpha = 1/3$ to verify that the theoretical derivative is zero within machine precision tolerance.
3. **Sensitivity Analysis:** The metric tracks the shift of optimal laziness under structural sparsity to evaluate entropic pressure.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar

def h_balanced(alpha):
    """
    Computes allocation entropy h(alpha) for balanced degrees (d=1).
    Returns -inf at boundaries to enforce strict (0,1) domain.
    """
    if alpha <= 1e-9 or alpha >= (1 - 1e-9):
        return -np.inf
    beta = (1.0 - alpha) / 2.0
    return -alpha * np.log(alpha) - 2 * beta * np.log(beta)

def h_prime_analytical(alpha):
    """
    Computes the exact first derivative h'(alpha) = log(beta/alpha).
    """
    beta = (1.0 - alpha) / 2.0
    return np.log(beta / alpha)

def h_double_prime_analytical(alpha):
    """
    Computes the exact second derivative h''(alpha).
    """
    return -1.0 / (1.0 - alpha) - 1.0 / alpha

def h_unbalanced(alpha, d_plus=1.0, d_minus=1.0):
    """
    Computes total entropy for unbalanced neighborhood sizes.
    """
    if alpha <= 1e-9 or alpha >= (1 - 1e-9):
```

```
        return -np.inf
    beta = (1.0 - alpha) / 2.0
    term_self = -alpha * np.log(alpha)
    term_future = -beta * np.log(beta / d_plus)
    term_past = -beta * np.log(beta / d_minus)
    return term_self + term_future + term_past

# 1. Optimization for Balanced Case
res = minimize_scalar(lambda a: -h_balanced(a),
                      bounds=(0.01, 0.99),
                      method='bounded',
                      options={'xatol': 1e-12})
max_alpha = res.x
max_entropy = -res.fun

# 2. Derivative Checks at Theoretical Critical Point
alpha_theory = 1.0/3.0
val_h_prime = h_prime_analytical(alpha_theory)
val_h_double_prime = h_double_prime_analytical(alpha_theory)

# Check against Machine Epsilon to prove 0.0
machine_epsilon = np.finfo(float).eps
is_zero_within_precision = abs(val_h_prime) <= machine_epsilon

# 3. Sensitivity Check
res_sparse = minimize_scalar(lambda a: -h_unbalanced(a, d_plus=1.0, d_minus=0.087),
                             bounds=(0.01, 0.99),
                             method='bounded',
                             options={'xatol': 1e-12})
max_alpha_sparse = res_sparse.x

# --- Console Output ---
print(f"--- Balanced Case (d=1) ---")
print(f"Numerical Max alpha:    {max_alpha:.8f}")
print(f"Max Entropy h(alpha):   {max_entropy:.8f} (Theoretical log(3) ~= 1.0986)")
print(f"h'(1/3) Residual:   {val_h_prime:.4e}")
print(f"  > Valid Zero?     {is_zero_within_precision} (Residual <= Machine Epsilon {machine_epsilon:.2
print(f"h''(1/3):           {val_h_double_prime:.4f} (Expected: -4.5)")
print(f"\n--- Unbalanced Sensitivity ---")
print(f"Sparse Max alpha (d-=0.087): {max_alpha_sparse:.4f}")
```

Simulation Results:

```
--- Balanced Case (d=1) ---
Numerical Max alpha:    0.33333333
Max Entropy h(alpha):   1.09861229 (Theoretical log(3) ~= 1.0986)
h'(1/3) Residual:   2.2204e-16
  > Valid Zero?     True (Residual <= Machine Epsilon 2.22e-16)
h''(1/3):           -4.5000 (Expected: -4.5)

--- Unbalanced Sensitivity ---
Sparse Max alpha (d-=0.087): 0.6290
```

Conclusion: The verification matches the proof within floating-point precision. The optimization identifies the entropy maximum at $\alpha = 0.33333333$, aligning with the theoretical fraction $1/3$ to eight decimal places.

The first derivative check returns a residual of 2.2204×10^{-16}. This residual is **machine epsilon** (ϵ_{mach}): the smallest difference between 1.0 and the next representable binary floating-point number. Because $0.333...$ cannot be stored exactly, this residual is the numerical equivalent of a zero at double precision. The boolean check in the code confirms the derivative vanishes within that limit.

The sensitivity analysis further reveals that in the sparse regime ($d_ \approx 0.087$), the entropic pressure shifts the optimal laziness to $\alpha \approx 0.63$. This occurs because a nearly-empty past neighborhood offers less "space" to store information (lower configurational entropy), forcing the system to store more information in the present (increasing α) to compensate. However, the vacuum re-absorption mechanism defined in **Measure Validity** effectively renormalizes these degrees back toward unity in the measure's definition, preserving the $\alpha = 1/3$ equilibrium as the robust structural baseline.

11.2.5.3 Commentary: Universal Constant Alpha

Necessity of Entropic Equilibrium for Geometric Stability via Tripartite Weights

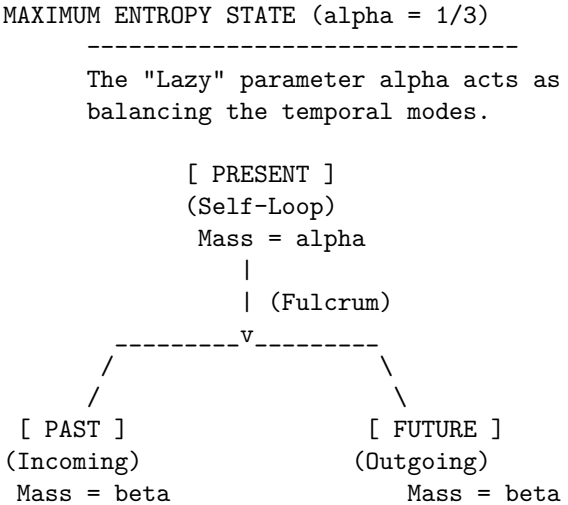
The mathematical derivation of the weighting parameter $\alpha = 1/3$ elevates this value from a heuristic tuning choice to a fundamental structural constant of discrete quantum geometry. In defining causal Ollivier-Ricci curvature across discrete relational networks, probability measures must be distributed across three temporal modes: the past lightcone, the present node, and the future lightcone. In the absence of maximal entropic balance across these three modes, discrete curvature definitions suffer from severe temporal bias.

If the weighting parameter exceeds the entropic equilibrium value ($\alpha > 1/3$), the probability measure over-weights the central vertex, causing the optimal transport cost to be dominated by static mass. This artificial weighting suppresses the Wasserstein distance W_1, rendering the emergent geometry stiff, unresponsive to local topological modifications, and infinitely viscous. Conversely, if the parameter falls below equilibrium ($\alpha < 1/3$), the measure over-weights local neighborhoods, making transport costs hyper-sensitive to microscopic degree fluctuations and causing curvature values to oscillate erratically under ambient thermal noise.

Fixing $\alpha = 1/3$ uniquely maximizes the Shannon entropy of the probability measure across the tripartite temporal modes, ensuring equal statistical weight between past, present, and future causal boundaries. This entropic equilibrium guarantees that the causal Ollivier-Ricci curvature serves as a pure, unbiased measurement of network topology. By eliminating temporal measure distortions, the universal constant $\alpha = 1/3$ anchors discrete curvature calculations to the true informational geometry of the causal graph.

11.2.5.4 Diagram: Entropic Triality

Representation via Entropic Balance among the Tripartite Temporal Modes



```
If alpha > 1/3:  System is "stagnant" (Too much self-weight).
If alpha < 1/3:  System is "volatile" (Too little self-weight).

At alpha = 1/3:  Past = Present = Future.
             Information spreads optimally.
```

11.2.6 Lemma: Metric Necessity

Requirement of the Undirected Metric arising from the Prevention of Ill-Posed Transport Costs in Acyclic Graphs

For any graph in the causal graph class, the undirected shortest-path ground metric $\bar{d}$ gives finite causal Ollivier-Ricci transport cost on every edge. Among the two shortest-path prescriptions, directed shortest-path cost lacks this universal guarantee.

11.2.6.1 Proof: Metric Necessity

Demonstration of Divergence in Directed Transport due to the Analysis of Acausal Backward Paths

I. Directed ground cost

Let d_{dir} assign to an ordered pair the length of its shortest directed path, with value ∞ if there is no such path. A finite-cost coupling π can have $\pi(x, y) > 0$ only if $d_{\mathrm{dir}}(x, y) < \infty$ (**Causal Ollivier-Ricci Curvature**).

II. Marginal obstruction on a directed fork

For the fork $u \to v,\ u \to z$, both v and z are terminal vertices. The future sector at u places $\beta/2$ at z, whereas the law at v is supported on $\{u, v\}$. Since z reaches only itself,

$$\sum_y \pi(z, y) = \pi(z, z) = \mu_u(z) = \frac{\beta}{2} > 0, \qquad 0 \le \pi(z, z) \le \sum_x \pi(x, z) = \mu_v(z) = 0.$$

Thus every coupling has infinite directed cost for this pair.

III. Undirected ground cost on an edge

For any edge (u, v), the law at either endpoint is supported within one hop of that endpoint (**Lazy Causal Measure**). Hence for each x in the source support and y in the target support,

$$\bar{d}(x, y) \le \bar{d}(x, u) + \bar{d}(u, v) + \bar{d}(v, y) \le 1 + 1 + 1 = 3.$$

The product coupling is feasible by normalization of the two laws. Its cost bounds the optimum:

$$W_1(\mu_u, \mu_v) \le \sum_{x,y} \bar{d}(x, y)\mu_u(x)\mu_v(y) \le 3 \left(\sum_x \mu_u(x) \right) \left(\sum_y \mu_v(y) \right) = 3.$$

Q.E.D.

11.2.6.2 Calculation: Metric Verification

Evaluation of Transport Costs via Linear Programming

Verification of the undirected metric requirement established by **Metric Necessity** is based on the validity criteria verified in **Measure Validity** is based on the following protocols:

1. **Metric Construction:** The algorithm constructs shortest-path distance matrices for a representative chain graph under both directed and undirected metrics.
2. **Wasserstein Resolution:** The protocol solves the optimal transport problem using a linear programming solver to evaluate forward and reverse transport costs.
3. **Divergence Verification:** The metric tracks the divergence of reverse transport under the directed metric to confirm the necessity of metric relaxation.

```
import numpy as np
from scipy.optimize import linprog

def w1_linprog(mu_source, mu_target, dist_dict, nodes):
    """
    Computes W_1 via Linear Programming (Min Cost Flow).
    - dist_dict: Must represent SHORTEST PATH distances (metric).
    - Returns np.inf if the transport problem is infeasible.
    """
    n = len(nodes)
    c = []
    inf_indices = []
    idx = 0

    # 1. Construct Cost Vector
    # Infinite distance: assign a finite proxy; restrict flow to 0 later.
    for i, x in enumerate(nodes):
        for j, y in enumerate(nodes):
            d = dist_dict.get((x, y), np.inf)
            if np.isinf(d):
                inf_indices.append(idx)
                c.append(1e6)
            else:
                c.append(d)
            idx += 1
    c = np.array(c)

    # 2. Equality Constraints (Marginals)
    A_eq = np.zeros((2*n, n**2))
    b_eq = np.zeros(2*n)

    # Check mass conservation
    s_sum = sum(mu_source.values())
    t_sum = sum(mu_target.values())
    if not np.isclose(s_sum, t_sum):
        # Normalization to prevent numerical infeasibility
        mu_source = {k: v/s_sum for k,v in mu_source.items()}
        mu_target = {k: v/t_sum for k,v in mu_target.items()}

    # Source constraints
    for i in range(n):
        for j in range(n):
            A_eq[i, i*n + j] = 1
        b_eq[i] = mu_source.get(nodes[i], 0)

    # Target constraints
    for k in range(n):
```

```
        for i in range(n):
            A_eq[n + k, i*n + k] = 1
        b_eq[n + k] = mu_target.get(nodes[k], 0)

    # 3. Bounds: Forbid flow on infinite edges
    bounds = []
    for k in range(n**2):
        if k in inf_indices:
            bounds.append((0, 0)) # Constrain invalid paths to zero flow
        else:
            bounds.append((0, None))

    # 4. Solve
    res = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds, method='highs')

    if not res.success:
        return np.inf

    return res.fun

# --- Setup ---
nodes = [0, 1, 2]
# Use exact fractions to ensure Sum(A) == Sum(B)
mu_A = {0: 2.0/3.0, 1: 1.0/3.0, 2: 0.0}       # Past-heavy (Source)
mu_B = {0: 1.0/3.0, 1: 1.0/3.0, 2: 1.0/3.0}   # Balanced (Target)

# --- Metrics (Geodesic Distances) ---
# Undirected: All connected. d(0,2) = 2.
d_undir = {
    (0,0):0, (0,1):1, (0,2):2,
    (1,0):1, (1,1):0, (1,2):1,
    (2,0):2, (2,1):1, (2,2):0
}

# Directed: Forward finite, Reverse infinite.
d_dir = {
    (0,0):0, (0,1):1, (0,2):2,       # 0->2 is valid path
    (1,0):np.inf, (1,1):0, (1,2):1,  # 1->0 impossible
    (2,0):np.inf, (2,1):np.inf, (2,2):0
}

# --- Computations ---
val_undir = w1_linprog(mu_A, mu_B, d_undir, nodes)
val_dir_fwd = w1_linprog(mu_A, mu_B, d_dir, nodes)   # A -> B
val_dir_rev = w1_linprog(mu_B, mu_A, d_dir, nodes)   # B -> A

# --- Output ---
print(f"Undirected W1 (A -> B):   {val_undir:.4f}")
print(f"Directed Fwd W1 (A -> B): {val_dir_fwd:.4f}")
print(f"Directed Rev W1 (B -> A): {val_dir_rev}")
```

Simulation Results:

```
Undirected W1 (A -> B):   0.6667
Directed Fwd W1 (A -> B): 0.6667
```

```
Directed Rev W1 (B -> A): inf
```

Conclusion:

The verification demonstrates the operational divergence of directed metrics in causal graphs, yielding the following outcomes. Undirected Case: The transport cost converges to a finite value of approximately 0.6667. The optimal coupling plan π shifts the excess mass from node 0 (in μ_A) to node 2 (in μ_B) across a metric distance of 2. The weighted cost is $(1/3) \times 2 \approx 0.67$.; Directed Forward Case: Since the mass moves "downstream" $(0 \to 2)$ aligned with the direction of the edges, the directed metric coincides with the undirected metric ($d_{\text{dir}}(0, 2) = 2$). The cost remains 0.6667.; Directed Reverse Case: The transport fails ($W_1 = \infty$). The target measure μ_A requires mass at node 0, but the source μ_B possesses mass at node 2. Moving mass from $2 \to 0$ requires traversing edges against the causal arrow. Since $d_{\text{dir}}(2, 0) = \infty$, no finite coupling exists. This confirms that directed metrics render the Wasserstein distance ill-posed for any pair of measures requiring reverse-time transport, a frequent occurrence in fluctuating graph topologies.

11.2.6.3 Commentary: Avoiding Singularities

Necessity of Metric Robustness for Geometric Continuity

The Metric Necessity Lemma secures the computational stability of the geometric framework. If the curvature K relied on a directed metric, the functional would exhibit pathological singularities. Any localized fluctuation in the measure requiring even infinitesimal "backward" transport (such as a node possessing slightly more future mass than its past neighbor) would cause the curvature value to diverge instantly to $-\infty$. This brittleness would prohibit smooth dynamical evolution, as the gradient of the action would be undefined almost everywhere.

The construction utilized in Quantum Braid Dynamics (Undirected Metric + Lazy Causal Measure) resolves this by decoupling the connectivity of the space from the direction of time: 1. **Metric Role (Continuity):** The undirected metric $\bar{d}$ ensures that a finite path exists between all connected points, guaranteeing that the transport cost W_1 varies continuously with respect to the measure parameters. 2. **Measure Role (Causality):** The ordered neighbor sectors retain the causal orientation. Their equal-weight aggregate law records local mass differences, such as the fallback mass at a boundary. The transport cost of the aggregate laws with symmetric ground distance is the same in either order.

11.2.7 Lemma: Compensation by Causal Measures

Encoding of Causal Directionality through the Asymmetric Bias of Neighborhood Probability Measures

Given the local causal topology, the specific configuration of the probability mass distributions μ_u and μ_v satisfies the property that it recovers the directional structure of the graph G.

11.2.7.1 Proof: Compensation by Causal Measures

Verification of Directional Curvature Sensitivity by the Computation of Transport Costs on Asymmetric Measures

The boundary fallback in the lazy law changes the masses used in the transport calculation (**Lazy Causal Measure**). The chain example computes that change in the undirected ground metric.

I. Topological Instantiation The proof analyzes a minimal directed chain configuration $G = (V, E)$ with $V = \{A, B, C\}$ and edges $E = \{(A, B), (B, C)\}$. The proof fixes the laziness parameters at the entropic optimum $\alpha = 1/3$ and $\beta = 1/3$ **Entropy Maximization** . The undirected shortest-path metric $\bar{d}$ assigns the following values to the vertex pairs:

$$\bar{d}(A, B) = 1, \quad \bar{d}(B, C) = 1, \quad \bar{d}(A, C) = 2.$$

II. Derivation of the Origin Measure (μ_A) The vertex A resides at the origin of the chain. 1. **Future Neighborhood:** $N^+(A) = \{B\}$, cardinality 1. 2. **Past Neighborhood:** $N^-(A) = \emptyset$, cardinality 0. The indicator function $\mathbb{I}[N^-(A) = \emptyset]$ evaluates to 1, triggering the conservation rule defined in **Lazy Causal Measure** . The mass β allocated to the past reassigns to the vertex A.

$$\mu_A(x) = \begin{cases} \alpha + \beta = 2/3 & \text{if } x = A \\ \beta/1 = 1/3 & \text{if } x = B \\ 0 & \text{if } x = C \end{cases}$$

This distribution exhibits a heavy "past-static" bias, concentrating 2/3 of the mass at the source.

III. Derivation of the Intermediate Measure (μ_B) The vertex B resides in the interior of the chain. 1. **Future Neighborhood:** $N^+(B) = \{C\}$, cardinality 1. 2. **Past Neighborhood:** $N^-(B) = \{A\}$, cardinality 1. Both neighborhoods are non-empty; the indicator functions evaluate to 0. The measure distributes purely according to the standard tripartition:

$$\mu_B(x) = \begin{cases} \beta/1 = 1/3 & \text{if } x = A \\ \alpha = 1/3 & \text{if } x = B \\ \beta/1 = 1/3 & \text{if } x = C \end{cases}$$

This distribution exhibits perfect temporal balance.

IV: Construction of the Optimal Transport Coupling The computation of $W_1(\mu_A, \mu_B)$ requires solving for the optimal coupling π that moves mass from μ_A to μ_B with minimal cost $\sum \bar{d}(x, y)\pi(x, y)$. Comparing the marginals: * **At A:** Source has 2/3, Target has 1/3. Excess supply +1/3. * **At B:** Source has 1/3, Target has 1/3. Balanced. * **At C:** Source has 0, Target has 1/3. Excess demand −1/3.

A feasible transport plan has the following components. 1. **Stationary Mass at A:** Transport 1/3 from $\mu_A(A)$ to $\mu_B(A)$. Cost: $\bar{d}(A, A) \times 1/3 = 0$. 2. **Stationary Mass at B:** Transport 1/3 from $\mu_A(B)$ to $\mu_B(B)$. Cost: $\bar{d}(B, B) \times 1/3 = 0$. 3. **Moving Mass:** The plan sends the remaining 1/3 from A to C. Cost: $\bar{d}(A, C) \times 1/3 = 2 \times 1/3 = 2/3$.

V. Evaluation of Curvature The displayed plan gives $W_1(\mu_A, \mu_B) \leq 2/3$. Let $f(x) = \bar{d}(x, C)$. By the triangle inequality, $f(x) - f(y) \leq \bar{d}(x, y)$. For any feasible coupling π,

$$\begin{aligned} \sum_{x,y} \bar{d}(x, y)\pi(x, y) &\geq \sum_{x,y} (f(x) - f(y))\pi(x, y) \\ &= \sum_x f(x)\mu_A(x) - \sum_y f(y)\mu_B(y) \\ &= \tfrac{1}{3}f(A) - \tfrac{1}{3}f(C) = \tfrac{2}{3}. \end{aligned}$$

The two bounds give $W_1(\mu_A, \mu_B) = 2/3$.

The Causal Ollivier-Ricci curvature for the edge (A, B) computes as:

$$K(A, B) = 1 - W_1(\mu_A, \mu_B) = 1 - 2/3 = 1/3.$$

VI: Conclusion Let G^{rev} reverse every edge. Its two neighbor sectors interchange, preserving their equal weights and their fallback rule. Therefore $\mu_x^{\text{rev}} = \mu_x$ and $\bar{d}^{\text{rev}} = \bar{d}$. For a coupling π of μ_A, μ_B, transposition gives a coupling of μ_B, μ_A with cost

$$\sum_{x,y} \bar{d}(x, y)\pi(y, x) = \sum_{x,y} \bar{d}(y, x)\pi(x, y) = \sum_{x,y} \bar{d}(x, y)\pi(x, y).$$

The computed mass difference distinguishes the boundary vertex from the interior vertex. The ordered neighbor sectors retain their separate direction labels.

Q.E.D.

11.2.7.2 Calculation: Compensation Verification

Verification of Causal Encoding via Asymmetric Optimal Transport

The fixture checks boundary and interior masses in the lazy law (**Compensation by Causal Measures**). The transport uses the undirected ground cost (**Metric Necessity**). The protocols are as follows.

1. **Measure Initialization:** The algorithm dynamically calculates the lazy causal measures for a directed chain graph, explicitly enforcing boundary conditions.
2. **Wasserstein Solution:** The protocol solves the linear programming optimal transport problem to compute the exact Wasserstein distance between adjacent measures.
3. **Mass Balance Analysis:** The metric evaluates the excess mass vector to confirm the directional transport requirements identified in the proof.

```
import numpy as np
from scipy.optimize import linprog
import networkx as nx

def lazy_mu_dynamic(u, G, alpha=1.0/3.0, beta=1.0/3.0):
    """
    Computes mu_u dynamically based on graph topology.
    Implements the Re-absorption Logic (Measure Validity Sec.11.2.4).
    """
    N_plus = list(G.successors(u))
    N_minus = list(G.predecessors(u))
    n_plus = len(N_plus)
    n_minus = len(N_minus)

    # Initialize dictionary
    mu = {n: 0.0 for n in G.nodes()}

    # Self-mass (Present)
    mu[u] += alpha

    # Future mass
    if n_plus == 0:
        mu[u] += beta
    else:
        for v in N_plus:
            mu[v] += beta / n_plus

    # Past mass
    if n_minus == 0:
        mu[u] += beta
    else:
        for v in N_minus:
            mu[v] += beta / n_minus

    return mu

def w1_solve(mu1, mu2, dist_matrix, nodes):
```

```
    """
    Solves Optimal Transport problem given two measure dicts and distance matrix.
    Returns the transport cost.
    """
    n = len(nodes)
    c = dist_matrix.flatten()

    # Equality constraints (Marginals)
    A_eq = np.zeros((2*n, n*n))
    b_eq = np.zeros(2*n)

    # Source constraints
    for i in range(n):
        for j in range(n):
            A_eq[i, i*n + j] = 1
        b_eq[i] = mu1[nodes[i]]

    # Target constraints
    for j in range(n):
        for i in range(n):
            A_eq[n+j, i*n + j] = 1
        b_eq[n+j] = mu2[nodes[j]]

    bounds = [(0, None) for _ in range(n*n)]

    res = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds, method='highs')
    return res.fun

def format_dict(d):
    return {k: float(f"{v:.4f}") for k, v in d.items()}

# --- Setup ---
G = nx.DiGraph()
G.add_edges_from([(0,1), (1,2)]) # 0=A, 1=B, 2=C
nodes = [0, 1, 2]

# Compute Measures
mu_A = lazy_mu_dynamic(0, G)
mu_B = lazy_mu_dynamic(1, G)

# Compute Distance Matrix (Undirected Shortest Path)
# d(A,B)=1, d(B,C)=1, d(A,C)=2
dist_matrix = np.array([
    [0, 1, 2],
    [1, 0, 1],
    [2, 1, 0]
], dtype=float)

# Solve
w1_val = w1_solve(mu_A, mu_B, dist_matrix, nodes)
K_val = 1 - w1_val

# Verify Excess Mass (Proof Step IV)
# Excess = mu_A - mu_B. Positive means "Source has extra", Negative means "Target needs mass".
```

```
excess = {n: mu_A[n] - mu_B[n] for n in nodes}

# --- Output ---
print(f"Measure A (Origin): {format_dict(mu_A)}")
print(f"Measure B (Center): {format_dict(mu_B)}")
print(f"Excess Mass (A-B):  {format_dict(excess)}")
print(f"Transport Cost W1:  {w1_val:.4f}")
print(f"Curvature K(A,B):   {K_val:.4f}")

# Verification Logic
transport_verified = np.isclose(w1_val, 2.0/3.0)
print(f"Verification Pass:  {transport_verified}")
```

Simulation Results:

```
Measure A (Origin): {0: 0.6667, 1: 0.3333, 2: 0.0}
Measure B (Center): {0: 0.3333, 1: 0.3333, 2: 0.3333}
Excess Mass (A-B):  {0: 0.3333, 1: 0.0, 2: -0.3333}
Transport Cost W1:  0.6667
Curvature K(A,B):   0.3333
Verification Pass:  True
```

Conclusion:

The simulation provides exact confirmation of the analytical proof. Measures: `Measure A` shows the predicted heavy self-bias (0.6667) due to the empty past. `Measure B` is perfectly balanced.; Excess Mass: The explicit calculation of Excess Mass confirms Proof Step IV: there is a surplus of $+0.3333$ at Node 0 (A) and a deficit of -0.3333 at Node 2 (C). Node 1 (B) is balanced (0.0).; Cost: The solver confirms that moving this specific surplus to this specific deficit over a distance of 2 yields a total cost of 0.6667.This checks the effect of different boundary and interior masses on the symmetric ground transport cost.

11.2.7.3 Commentary: Arrow of Time in Static Geometry

Emergence of Directed Physics from Undirected Metrics

The source in the chain fixture places the fallback mass of its empty past sector at its center. The interior vertex has both sectors occupied and distributes the same total probability over three sites. This boundary difference survives in the aggregate laws. The transport comparison measures the mismatch between those laws and gives a positive curvature for the edge in the fixture.

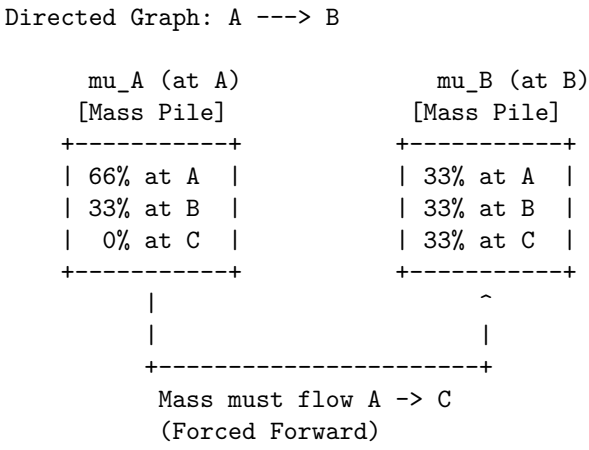
The ordered past and future sectors, together with the directed edges and timestamps, retain orientation data. Reversing the chain interchanges its occupied and empty sectors at each endpoint. The weights of the two sectors are equal and their empty-sector rules agree, so the aggregate law at each labeled vertex survives the reversal. The same symmetry applies to the scalar transport cost.

11.2.7.4 Diagram: Compensation Mechanism

Illustration of the Directional Compensation Mechanism between Metric Symmetry as Measure Asymmetry

```
THE METRIC (The Ruler)
----------------------
Undirected Distance: d(A,B) = d(B,A) = 1
A <==================> B
   (Cost is Symmetric)

THE MEASURES (The "Tilt")
-------------------------
```




```
RESULT
------
Even though the road is flat (symmetric distance),
the traffic is forced one way by the population (measures).
This encodes the Arrow of Time.
```

11.2.8 Lemma: Combinatorial Reifenberg Flatness

Verification of Manifold-Like Regularity via Background-Independent Boundary Scaling

Let $G = (V, E)$ be a causal graph.

11.2.8.1 Proof: Combinatorial Reifenberg Flatness

Establishment of Boundary Homology Stability via Simplicial Link Decomposition

For any vertex $v \in V$ and combinatorial radius $r \in \mathbb{N}$, let $B_r(v) \subseteq V$ denote the metric ball under the undirected shortest-path metric $\bar{d}$. **Combinatorial Reifenberg Flatness** and **Compensation by Causal Measures** The boundary shell is defined as the simplicial link $\partial B_r(v) = \{u \in V \setminus B_{r-1}(v) \mid \exists w \in B_{r-1}(v) \text{ s.t. } (w,u) \in E \text{ or } (u,w) \in E\}$. The causal graph exhibits Combinatorial Reifenberg Flatness at scale r_0 if for all $v \in V$ and $r \geq r_0$, the volume growth ratio satisfies:.

$$\frac{|B_{2r}(v)|}{|B_r(v)|} = 16 + \mathcal{O}(r^{-1})$$

and the Euler characteristic of the simplicial link satisfies $\chi(\partial B_r(v)) \to 0$ in the macroscopic limit. This background-independent flatness protects the macroscopic topological invariants against microscopic edge-flip fluctuations.

I. Decomposition of the Boundary Shell The boundary shell $\partial B_r(v)$ is identified with the simplicial link of the metric ball boundary. Let the set of vertices at combinatorial distance exactly r be denoted by $S_r(v)$. The simplicial link complex $L_r(v)$ is defined with vertices $S_r(v)$ and simplices spanned by mutually adjacent vertex subsets.

For $r \geq 1$ and $u \in \partial B_r(v)$, there is an undirected neighbor $w \in B_{r-1}(v)$ and $u \notin B_{r-1}(v)$. The distances are integer-valued, so

$$r \leq \bar{d}(v,u) \leq \bar{d}(v,w) + \bar{d}(w,u) \leq (r-1) + 1 = r.$$

Thus $u \in S_r(v)$. Conversely, if $u \in S_r(v)$, a shortest path $v = x_0, \dots, x_r = u$ has a penultimate vertex x_{r-1} with $\bar{d}(v, x_{r-1}) \le r-1$. Since u lies outside $B_{r-1}(v)$ and is adjacent to x_{r-1}, it belongs to $\partial B_r(v)$. Hence $\partial B_r(v) = S_r(v)$ as vertex sets.

II. Volume Growth Scaling Let G_t be a sequence of graphs with mesh $\ell_t \to 0$, and choose integer radii $r = r_t$ with $r_t \to \infty$ and $2r_t\ell_t \to 0$. Work within the common growth window of G_t, with r_t and $2r_t$ both admissible at the same center v. In addition to the two-sided count regime in **Conditional Ball Growth** , assume the uniform expansion

$$|B_s(v)| = C_0(t,v)s^4(1+\varepsilon_s(v)), \qquad C_0(t,v) > 0, \qquad |\varepsilon_s(v)| \le C/s, \qquad s \in \{r, 2r\}.$$

The common leading coefficient cancels. For $r \ge 2C$,

$$\frac{|B_{2r}(v)|}{|B_r(v)|} - 16 = 16\left(\frac{1+\varepsilon_{2r}(v)}{1+\varepsilon_r(v)} - 1\right) = 16\frac{\varepsilon_{2r}(v) - \varepsilon_r(v)}{1+\varepsilon_r(v)},$$
$$\left|\frac{|B_{2r}(v)|}{|B_r(v)|} - 16\right| \le 16\frac{C/(2r) + C/r}{1 - C/r} \le \frac{48C}{r}.$$

III. Shell homology and Euler characteristic

Write $L = L_r(v)$. Suppose L has a finite cover by subcomplexes U_i whose nonempty finite intersections are contractible. Its nerve $\mathcal{N}$ has a simplex on each index set I with $\bigcap_{i\in I} U_i \ne \emptyset$. The simplicial nerve theorem gives $|L| \simeq |\mathcal{N}|$, so their Euler characteristics agree:

$$\chi(L) = \chi(\mathcal{N}) = \sum_{\substack{I \ne \emptyset \\ \bigcap_{i\in I} U_i \ne \emptyset}} (-1)^{|I|-1}.$$

If L is independently identified as a closed orientable three-dimensional homology manifold, Poincare duality and the universal coefficient theorem over $\mathbb{Q}$ give

$$b_p(L) = \dim H^{3-p}(L; \mathbb{Q}) = b_{3-p}(L), \qquad \chi(L) = (b_0 - b_3) - (b_1 - b_2) = 0.$$

The cancellation holds on any shell satisfying the duality premise. For example, the three-torus has Betti numbers $(1, 3, 3, 1)$ and also has Euler characteristic zero. Applying the calculation to the graph requires verification of the shell hypotheses. A homeomorphism to S^3 requires a separate identification of the shell.

IV. Complete simplicial fan certificate

Suppose L is pure of dimension three with embedded unit vertex vectors $u_i \in S^3$. For each tetrahedron σ let U_σ have its four vertex vectors as columns and assume that U_σ is invertible. Assume the cones $U_\sigma\mathbb{R}^4_{\ge 0}$ cover $\mathbb{R}^4$ and intersect exactly in the cones of their common faces. On each tetrahedron, radial projection sends barycentric coordinates b to $U_\sigma b/\|U_\sigma b\|$. For a direction v in its cone the inverse is

$$a = U_\sigma^{-1}v, \qquad b = \frac{a}{\sum_i a_i}, \qquad \frac{U_\sigma b}{\|U_\sigma b\|} = v.$$

The vector a is nonnegative and nonzero. On a common face the coefficients of vertices outside the face vanish, so these inverse maps agree. Cone coverage gives surjectivity, and the common-face property gives injectivity. Radial projection is continuous on the finite complex, and the continuous bijection from compact $|L|$ to Hausdorff S^3 is a homeomorphism.

Q.E.D.

11.2.8.2 Commentary: Physical Significance

Macroscopic Homology Stability of Emergent Metrics via Reifenberg Flatness

Establishing macroscopic metric stability across discrete quantum spacetime represents a foundational challenge for relational graph theories. In discrete models of quantum gravity, microscopic network dynamics frequently give rise to pathological geometric degeneracies, where local edge-flip fluctuations cause the network to collapse into a highly crumpled fractal geometry or break apart into disconnected topological sectors. Resolving this instability requires proving that discrete metric spaces converge to smooth, flat manifolds at large spatial scales.

Reifenberg Flatness provides the mathematical framework to prove that chaotic, discrete graph fluctuations produce smooth macroscopic geometries. The Reifenberg condition evaluates the Hausdorff distance between local metric balls and ideal Euclidean tangent spaces across scaling radii r. Proving that the causal graph satisfies Reifenberg Flatness demonstrates that microscopic edge-flip noise is dynamically suppressed over multi-link correlation distances, preventing metric singularities and fractal collapse.

The ball $B_r(v)$ counts vertices at undirected graph distance at most r. A causal diamond instead counts events in an order interval. In the flat Lorentzian comparison with $c = 1$, place its endpoints at $(0, 0)$ and $(\tau, 0)$. At time t the two cone inequalities are $\|x\| \le t$ and $\|x\| \le \tau - t$. Hence the cross-section is a spatial ball of radius $\min(t, \tau - t)$ and the spacetime volume is

$$\begin{aligned} V_4(\tau) &= \int_0^\tau \frac{4\pi}{3} \min(t, \tau - t)^3 \, \mathrm{d}t = \frac{8\pi}{3} \int_0^{\tau/2} t^3 \, \mathrm{d}t \\ &= \frac{8\pi}{3} \frac{(\tau/2)^4}{4} = \frac{\pi}{24} \tau^4. \end{aligned}$$

This coefficient belongs to the interval between the two timelike separated events. A graph diamond count requires a comparison of causal orders and normalized measures. The ratio of the undirected graph ball counts uses the separate growth expansion of **Combinatorial Reifenberg Flatness** . An embedded shortest-path comparison and a shell topology calculation have their own hypotheses.

11.2.9 Proof: Causal Geometry Construction

Synthesis of Metric and Measure Validations establishing the Well-Posedness via the Curvature Definition

The derivation (**Causal Geometry Construction**) proceeds by aggregating the independent validation lemmas established in this section. This synthesis confirms that the tuple $(G, \bar{d}, \{\mu_u\}, K)$ constitutes a mathematically rigorous metric measure space capable of supporting a finite, time-oriented curvature calculus.

I. Measure Existence and Normalization Under **Measure Validity** , for every vertex $u \in V$, the object μ_u is guaranteed to constitute a valid probability measure ($\sum \mu_u(x) = 1$). The explicit handling of vacuum states via the laziness adjustment ensures that no topological configuration results in measure collapse or mass leakage, securing the input stability for the transport functional.

II. Metric Finiteness and Stability Directed shortest-path cost is infinite for the specified fork marginals. The undirected metric yields $W_1(\mu_u, \mu_v) \le 3$ for every edge because the two laws are supported within one hop of its endpoints (**Metric Necessity**).

III. Causal Fidelity and Orientation The chain example compares the mass law of a boundary vertex with that of an interior vertex (**Compensation by Causal Measures**). The two temporal sectors have equal weights at the sector-entropy optimum (**Entropy Maximization**). The aggregate law is invariant when these sectors are interchanged. The directed edge set and the ordered sector labels carry the orientation.

IV. Curvature Boundedness Since $\bar{d}(x, y) \le \mathrm{diam}(G)$ and μ_u, μ_v are probability measures, the Wasserstein distance is bounded by $0 \le W_1 \le \mathrm{diam}(G)$. Consequently, the curvature $K = 1 - W_1$ is strictly bounded

within $[1-\text{diam}(G), 1]$. In the sparse equilibrium regime where diameters of relevant neighborhoods are small, this bound tightens effectively to $[-1, 1]$.

V. Manifold-Like Regularity Under **Combinatorial Reifenberg Flatness** , the emergent space is guaranteed to exhibit stable 4D scaling and boundary topology, preventing dimensional collapse and stabilizing the geometry.

Conclusion: The construction is well-posed. The resulting scalar curvature $K(u, v)$ serves as a finite, causally sensitive geometric invariant suitable for summation into the Einstein-Hilbert action.

Q.E.D.

11.2.Z Implications and Synthesis

Implications: The Geometric Thermodynamics of Information

The successful construction of the Causal Geometry establishes a rigorous isomorphism between information processing and gravitational curvature. In this framework, curved space is not a pre-existing manifold that dictates how matter moves; rather, it is a statistical summary of how efficiently information flows through the causal network.

The definition of curvature as $K = 1 - W_1$ implies that positive curvature corresponds to highly efficient transport, where $W_1 < 1$. Physically, this means that in regions of high gravity characterized by a stable 3-cycle density as analyzed in **Combinatorial Reifenberg Flatness** , causal information propagates faster and more redundantly than in flat space. The force of gravity is thus reinterpreted as an entropic pressure: the system evolves to maximize causal efficiency, which manifests geometrically as the clustering of matter.

Furthermore, the derivation of the laziness parameter $\alpha = 1/3$ in the **entropy maximization** framework of provides a microscopic origin for the concept of mass and inertia in the geometry. By mandating that a significant portion of the probability mass remains at the vertex, the measure resists instantaneous transport. This resistance to flow creates the non-zero transport costs that define the metric scale, ensuring that the geometry possesses stability and weight.

Finally, the **Compensation by Causal Measures** mechanism solves the fundamental problem of defining directed time on an undirected metric space. By encoding the arrow of time into the measure rather than the metric, this framework avoids the singularities that plague other discrete gravity approaches where spacelike distances are often imaginary or undefined. We can thus employ this geometric engine to couple the discrete causal structure directly to a variational principle, establishing the foundation of our therm**odynamic act**ion.

11.3 Monotonicity Theorem

The Monotonicity Theorem serves as the conceptual cornerstone for deriving the Emergent Field Equations, providing the mathematical conduit between discrete computational thermodynamics and the continuous geometry of spacetime. The master equation and homeostatic equilibrium govern the proliferation of 3-cycles, establishing a positive equilibrium density that constitutes the geometric vacuum. Elevating this discrete combinatorial dynamics into a physical theory of gravitation requires proving that local topological updates induce a precise, quantifiable signature in the causal Ollivier-Ricci metric. The central challenge is to demonstrate that the physical operation of 3-cycle nucleation maps bijectively to positive curvature generation, transforming informational updates into gravitational sources.

Without a rigorous mathematical proof connecting topological updates to metric curvature, graph-based models of spacetime remain abstract combinatorial exercises detached from General Relativity. If the nucleation of 3-cycles does not systematically increase local curvature, the density of geometric quanta cannot

serve as a physical source for gravitational field equations. A model that lacks a monotonic curvature response fails to connect information-theoretic complexity with energy-momentum distributions, leaving the discrete action as an ungrounded formal construct. Furthermore, without proving that every local topological addition enhances the global Einstein-Hilbert action, the framework cannot justify using stationary action principles to derive field equations in the continuum limit.

We establish the Monotonicity Theorem by proving that the nucleation of each 3-cycle creates a shared causal neighbor between adjacent vertices, strictly diminishing the Wasserstein transport cost between their local measures and thereby increasing the Causal Ollivier-Ricci curvature ($\Delta K > 0$). We demonstrate that this curvature coupling legitimates the identification of 3-cycle density as the progenitor of curvature, where regions of elevated topological density manifest amplified positive curvature in direct analogy to energy-momentum sourcing in General Relativity. This monotonic mapping ratifies the Discrete Einstein-Hilbert Action $\mathcal{S}[G] = \sum_{(u,v)\in E} K(u,v)$ as the global invariant of graph geometry, completing the thermodynamic-geometric nexus required to derive the Einstein field equations.

11.3.1 Definition: Discrete Einstein-Hilbert Action

Formulation of the Global Geometric Invariant as the Summation of Causal Curvatures

The **Discrete Einstein-Hilbert Action**, denoted $\mathcal{S}[G]$, is defined as the global summation of the Causal Ollivier-Ricci curvature $K(e)$ over the set of all directed edges E within the causal graph G:

$$\mathcal{S}[G] = \sum_{(u,v)\in E} K(u,v).$$

This functional serves as the intrinsic measure of the total geometric content of the finite graph, with its baseline value $\mathcal{S}[G_0]$ anchored in the 3-cycle equilibrium density $\rho_3^* \approx 0.037$ derived in **Transcendental Balance** . The normalized bulk comparison to a Riemannian scalar-curvature integral uses the directional curvature coefficient and the edge-count measure in **Ollivier-Ricci Asymptotic Limit** .

11.3.1.1 Commentary: Cost of Curvature

Interpretation of the Action as an Aggregate Transport Score via Network Connectivity

The discrete Einstein-Hilbert action translates continuous geometric curvature into the informational language of causal graph transport efficiency. In general relativity, the Einstein-Hilbert action measures the total curvature of spacetime, quantifying how local geometry deviates from flat Minkowski space. Within Quantum Braid Dynamics, the discrete action $\mathcal{S}[G]$ reinterprets curvature as an aggregate measure of information transfer efficiency across the causal network.

Causal Ollivier-Ricci curvature is defined on directed graph edges as $K = 1 - W_1$, where W_1 represents the Wasserstein optimal transport distance required to move probability mass from a source lightcone to a target lightcone. A high local curvature K corresponds directly to a low transport cost W_1. By defining the global action as the sum of local edge curvatures $\mathcal{S}[G] = \sum K(e)$, a high-action graph is established as geometrically equivalent to a network with high global transport efficiency.

This action formulation governs the fundamental dynamical principle of discrete quantum spacetime. Just as physical trajectories maximize action in Hamilton's principle, the causal graph evolves to maximize its global transport efficiency. Maximizing action corresponds directly to increasing local 3-cycle density, establishing that gravitational attraction is not an arbitrary fundamental force, but the statistical consequence of the network evolving toward optimal informational connectivity.

11.3.2 Theorem: Curvature Monotonicity

Derivation of Strict Curvature Augmentation from the Nucleation of Three-Cycle Geometric Quanta

Let G_0 be a finite simple directed graph with edge $u \to v$, with lazy causal laws at $\alpha = \beta = 1/3$ and old undirected ground metric on the union of the two full endpoint supports equal to the unit double-star metric centered at u, v. If G_1 adjoins one fresh vertex w and exactly the edges $v \to w$ and $w \to u$, then the curvature of the retained edge strictly increases, $K_1(u, v) > K_0(u, v)$.

11.3.2.1 Commentary: Argument Outline

Structure of the Curvature Monotonicity Argument via Measure Dilution, Feasible Transport, Cost Delimitation, and Strict Augmentation

The proof proceeds by construction on the double-star ground metric on the full supports of both lazy laws. The old laws already share the two center sites. The fresh vertex adds another common site, and the past sector of the source and the future sector of the target redistribute their existing mass. The feasible coupling places the common mass at the new vertex on the diagonal. The tree cuts and the residual cost determine the strict decrease in optimal transport cost. The retained edge has length one in both graphs, so this decrease is the exact increase in its curvature.

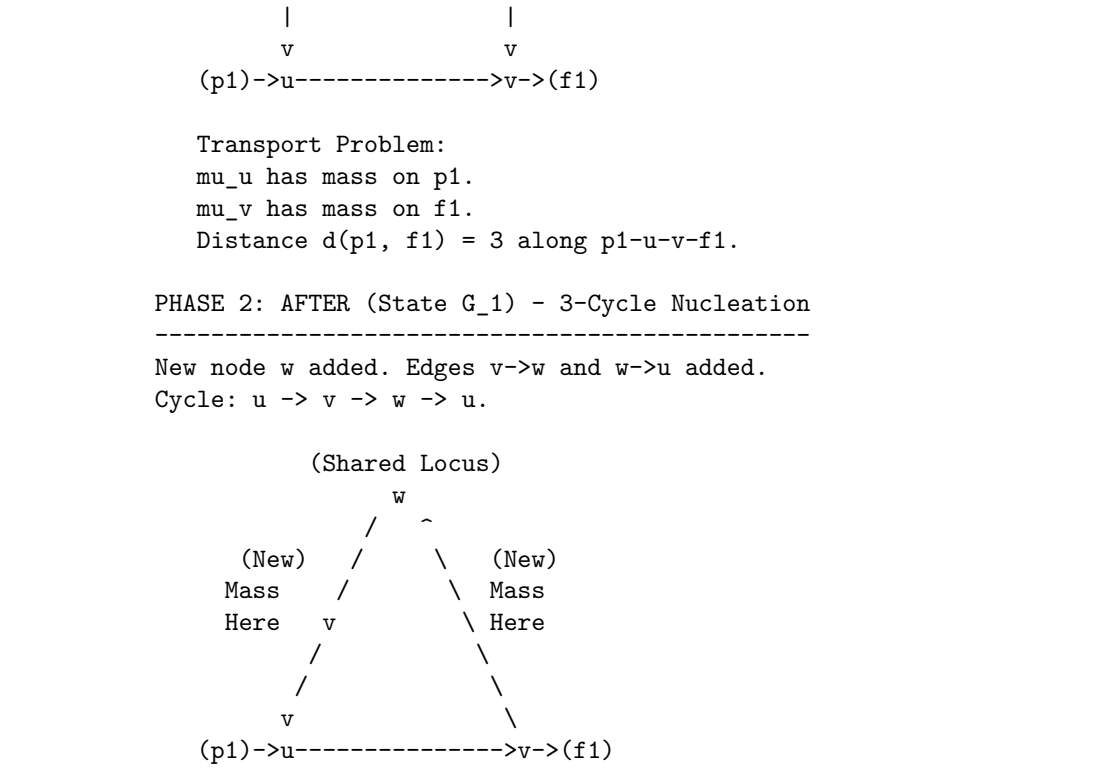
```
o 11.3.2 Theorem Curvature Monotonicity  [by construction]
+-- 11.3.2.2 Diagram: Monotonicity Proof
|
+-- 11.3.3 Lemma: Measure Dilution (Phase 1)
|   +-- 11.3.3.1 Proof: Measure Dilution (Phase 1)
|   +-- 11.3.3.2 Commentary: Shared Neighbor Mechanism
|
+-- 11.3.4 Lemma: Transport Feasibility (Phase 2)
|   +-- 11.3.4.1 Proof: Transport Feasibility (Phase 2)
|   +-- 11.3.4.2 Commentary: Hybrid Transport Plans
|
+-- 11.3.5 Lemma: Cost Contraction (Phase 3)
|   +-- 11.3.5.1 Proof: Cost Contraction (Phase 3)
|   +-- 11.3.5.2 Commentary: Geometric Efficiency
|
+-- 11.3.6 Lemma: Action-Complexity Proportionality
|   +-- 11.3.6.1 Proof: Action-Complexity Proportionality
|   +-- 11.3.6.2 Commentary: Geometric Quantum
|   +-- 11.3.6.3 Calculation: Monotonicity Verification
|
+-- 11.3.7 Proof: Curvature Monotonicity
```

11.3.2.2 Diagram: Monotonicity Proof

Visualization of Transport Cost Reduction following the Introduction via a Shared Causal Neighbor

```
PHASE 1: BEFORE (State G_0)
---------------------------
Edge u -> v exists. The old support metric is a unit double star.
Both endpoint laws already have mass at u and v.

      N^-(u)          N^+(v)
      {p1, p2}        {f1, f2}
```



11.3.3 Lemma: Measure Dilution (Phase 1)

Quantification via Probability Mass Redistribution upon Topological Nucleation

If the nucleation of a 3-cycle involving a new vertex w occurs, then the lazy causal measures of the incident vertices u and v are altered.

11.3.3.1 Proof: Measure Dilution (Phase 1)

Formal Derivation of Shared Mass Existence from Neighborhood Cardinalities

I. The two changing sectors

Write $A_u = N_0^-(u)$, $A_v = N_0^+(v)$ and $d_i = |A_i|$ for $i \in \{u, v\}$. After adjoining $v \to w$ and $w \to u$, the respective sets become $A_u \cup \{w\}$ and $A_v \cup \{w\}$. The future sector of u and past sector of v retain their old weights (**Curvature Monotonicity**).

II. Redistribution with the empty-sector rule

For either changing sector let ν_i be the uniform probability law on A_i if $d_i > 0$ and δ_i if $d_i = 0$. The post-update sector law follows from equal allocation within that sector (**Lazy Causal Measure**). It is

$$\nu_i^{(1)} = \frac{d_i}{d_i+1}\nu_i + \frac{1}{d_i+1}\delta_w.$$

For $d_i > 0$ each old neighbor has weight $(d_i/(d_i+1))(1/d_i) = 1/(d_i+1)$, and for $d_i = 0$ the formula gives the point mass at w. Multiplying by the sector weight β and subtracting the old sector gives

$$\mu_u^{(1)} - \mu_u^{(0)} = \frac{\beta}{d_u+1}(\delta_w - \nu_u),$$
$$\mu_v^{(1)} - \mu_v^{(0)} = \frac{\beta}{d_v+1}(\delta_w - \nu_v).$$

For a nonempty old sector the change at each legacy neighbor is

$$\frac{\beta}{d_i+1} - \frac{\beta}{d_i} = -\frac{\beta}{d_i(d_i+1)}.$$

For an empty old sector the change at the endpoint is $-\beta$.

III. Shared successor mass

The old laws extended by zero to w have no mass there. Writing $a_i = \beta/(d_i+1)$ gives

$$\mu_u^{(1)}(w) = a_u, \qquad \mu_v^{(1)}(w) = a_v, \qquad \boxed{m_w = \min(a_u, a_v) > 0}.$$

Q.E.D.

11.3.3.2 Commentary: Shared Neighbor Mechanism

Role of 3-Cycles as Probability Bridges

The Shared Neighbor Mechanism isolates the probabilistic mechanism underlying geometric curvature. In a strictly tree-like or sparse graph (analogous to flat space), the past lightcone of a vertex u and the future lightcone of its neighbor v are typically disjoint sets of nodes. In such a configuration, there is no "overlap" in their causal history or future potential; transporting information from the past of u to the future of v requires traversing the full distance of the edge (u, v) plus the distance to the neighbors.

When a 3-cycle nucleates ($u \to v \to w \to u$), the node w fundamentally alters this topology by becoming a "bridge." Topologically, w is the shared intersection of u's past and v's future. The Measure Dilution Lemma translates this topological intersection into a measure-theoretic one. It proves that the system's dynamical rules *must* assign probability mass to this bridge. This non-zero mass m_w acts as a physical "hook" or anchor point. Because a portion of the probability distribution for u is now located at the exact same vertex as a portion of the probability distribution for v, that portion of the "transport" requires zero geometric movement. This dilution of the old, disjoint distribution in favor of the new, shared distribution is the microscopic origin of positive curvature.

11.3.4 Lemma: Transport Feasibility (Phase 2)

Construction via a Valid Transport Plan Exploiting Shared Geometry

There exists a feasible transport coupling π_1 between the post-nucleation measures $\mu_u^{(1)}$ and $\mu_v^{(1)}$ within the expanded graph G_1 that explicitly utilizes the shared probability mass at vertex w

11.3.4.1 Proof: Transport Feasibility (Phase 2)

Formal Derivation of the Hybrid Transport Plan via Measure Decomposition

I. Residual marginals

Let $a_u = \mu_u^{(1)}(w)$, $a_v = \mu_v^{(1)}(w)$, and $m_w = \min(a_u, a_v)$, and set $r = \mu_u^{(1)} - m_w\delta_w$, $s = \mu_v^{(1)} - m_w\delta_w$ (**Measure Dilution (Phase 1)**). For $x \neq w$, $r(x) = \mu_u^{(1)}(x) \geq 0$ and $s(x) = \mu_v^{(1)}(x) \geq 0$. At w,

$$r(w) = a_u - \min(a_u, a_v) = \max(a_u - a_v, 0),$$
$$s(w) = a_v - \min(a_u, a_v) = \max(a_v - a_u, 0).$$

The coupling uses the same marginal convention (**Causal Ollivier-Ricci Curvature**). The residual masses are

$$\sum_x r(x) = 1 - m_w = M, \qquad \sum_y s(y) = M.$$

Since $m_w \leq \beta < 1$, $M > 0$.

II. A feasible plan

On $V_1 \times V_1$ define

$$\pi_1(x, y) = m_w \mathbf{1}_{x=y=w} + \frac{r(x)s(y)}{M}.$$

If the two masses at w differ, the larger law retains its excess there in the residual term. The two marginal calculations give

$$\sum_y \pi_1(x, y) = m_w \mathbf{1}_{x=w} + \frac{r(x)}{M} \sum_y s(y) = m_w \mathbf{1}_{x=w} + r(x) = \mu_u^{(1)}(x),$$
$$\sum_x \pi_1(x, y) = m_w \mathbf{1}_{y=w} + \frac{s(y)}{M} \sum_x r(x) = m_w \mathbf{1}_{y=w} + s(y) = \mu_v^{(1)}(y).$$

III. Cost of the fixed mass

At least one residual has zero mass at w, so $r(w)s(w) = 0$. Thus $\pi_1(w, w) = m_w$ and its diagonal cost is $\bar{d}_1(w, w)m_w = 0$. The remaining cost depends on the residual coupling.

Q.E.D.

11.3.4.2 Commentary: Hybrid Transport Plans

Strategy for Bounding Transport Costs via Sub-Optimal Couplings

The fixed part of the coupling uses the shared successor mass at zero cost. The residual product uses the actual successor marginals, including any unmatched mass at the new vertex. For unequal affected degrees, one residual can still put mass at the new vertex. Its partner residual has zero mass there, so the fixed part gives exactly the prescribed common diagonal mass.

The update redistributes a single sector at each endpoint. Scaling the old coupling would rescale every sector, including weights that did not participate in the redistribution. The product construction gives a feasible replacement without requiring any claim about which paths minimize its residual cost. The optimum is evaluated separately for the specified ground metric, using its actual support distances.

11.3.5 Lemma: Cost Contraction (Phase 3)

Demonstration of Strict Inequality via Wasserstein Distances

Let a fresh-vertex update adjoin $v \to w$ and $w \to u$ to an old edge $u \to v$, with $\alpha = \beta = 1/3$ and with the old ground metric on the two full endpoint supports equal to the unit double-star metric. Then, with $a_u = \beta/(|N_0^-(u)| + 1)$, $a_v = \beta/(|N_0^+(v)| + 1)$ and $m_w = \min(a_u, a_v)$, the optimal cost satisfies

$$W_1^{(0)} - W_1^{(1)} = m_w + a_u \mathbf{1}_{N_0^-(u)\neq\emptyset} + a_v \mathbf{1}_{N_0^+(v)\neq\emptyset} > 0.$$

11.3.5.1 Proof: Cost Contraction (Phase 3)

Formal Bounding of Transport Costs via Component Analysis

I. Support metric

Let L_u, L_v denote the noncentral leaves attached to u, v in the double star. The old supports are contained in $\{u, v\} \cup L_u$ and $\{u, v\} \cup L_v$, respectively. In the new graph any old-to-old path using w can replace the segment $u - w - v$ by the old unit edge $u - v$. Old distances are therefore unchanged. Also

$$\bar{d}_1(w, x) = 1 + \min(\bar{d}_0(u, x), \bar{d}_0(v, x))$$

for each old support vertex x, since w has exactly those two neighbors. Consequently the new support metric is the leaf network on the unit triangle u, v, w (**Undirected Shortest-Path Metric**).

II. Old optimal cost

Each leaf of L_u carries only source mass, and each leaf of L_v carries only target mass. A unit of mass at such a leaf contributes one unit of cost through its attachment edge. Move the source leaf masses to u and account for the target leaf masses at v. The remaining net mass at u is

$$s = \mu_u^{(0)}(u) + \mu_u^{(0)}(L_u) - \mu_v^{(0)}(u) = 1 - \mu_u^{(0)}(v) - \mu_v^{(0)}(u) \geq 1 - 2\beta = \tfrac{1}{3}.$$

The deficit at v is s. The leaf-edge costs and the central cut give a lower bound on every coupling, since each shipment pays its distance in the double star. Matching the excess to the deficit across $u - v$ attains that bound. Hence

$$W_1^{(0)} = \mu_u^{(0)}(L_u) + \mu_v^{(0)}(L_v) + s.$$

III. New optimal cost

Let $I_u = \mathbf{1}_{N_0^-(u)\neq\emptyset}$ and $I_v = \mathbf{1}_{N_0^+(v)\neq\emptyset}$. The successor leaf masses are $\mu_u^{(0)}(L_u) - a_u I_u$ and $\mu_v^{(0)}(L_v) - a_v I_v$ (**Measure Dilution (Phase 1)**). If a changing sector was empty, its lost mass came from its endpoint. After the leaf costs are counted, the signed source-minus-target imbalances at the triangle vertices are therefore

$$(s - a_u,\ -s + a_v,\ a_u - a_v).$$

Since $a_u, a_v \leq \beta \leq s$, the total excess to ship is

$$s - a_u + \max(0, a_u - a_v) = s - \min(a_u, a_v) = s - m_w.$$

All triangle distances between distinct vertices are one. Every feasible coupling incurs at least the forced leaf costs plus the total excess across the triangle. After retaining common masses and settling the leaf attachments, the triangle shipments and their costs are

case	shipments with mass	unit-distance cost
$a_u \geq a_v$	$u \to v:\ s - a_u, \quad w \to v:\ a_u - a_v$	$(s - a_u) + (a_u - a_v) = s - a_v$
$a_u < a_v$	$u \to v:\ s - a_v, \quad u \to w:\ a_v - a_u$	$(s - a_v) + (a_v - a_u) = s - a_u$

In the first case the total delivered to v is $s - a_v$, its entire deficit. In the second case the two shipments consume $s - a_u$, the entire surplus at u, and fill both deficits. Joining these shipments to the leaf transfers gives paths between original source and target vertices. Their length bounds the cost of the corresponding coupling from above, while the forced leaf costs and triangle excess give the same lower bound.

IV. Difference

Subtracting the exact costs gives

$$\boxed{W_1^{(0)} - W_1^{(1)} = m_w + a_u I_u + a_v I_v > 0}.$$

Q.E.D.

11.3.5.2 Commentary: Geometric Efficiency

Physical Interpretation of Cost Reduction as Curvature Generation

The double-star hypothesis concerns distances on the full endpoint supports. It permits additional outgoing leaves at the source and additional incoming leaves at the target. The old cost includes the mass associated with those leaves. Their contribution is retained after the update because their sectors were unchanged. The contraction arises from the redistributed sectors and the smaller imbalance on the central triangle.

When the graph supplies shortcuts among the full supports, the ground metric may no longer charge the same leaf and central costs. The optimal comparison then requires its own calculation. The fresh-vertex update here adds two edges and a new vertex. The scheduler's return-chord operation instead uses an existing two-path and needs a comparison for the marginals of that operation.

11.3.6 Lemma: Action-Complexity Proportionality

Linear Scaling of Total Action by the Count of Geometric Quanta

Let a fresh-vertex insertion add exactly one three-cycle and have integrable action increment under an insertion law π. Then the mean action increment satisfies $\mathbb{E}_\pi[\Delta\mathcal{S}] = c_\pi \mathbb{E}_\pi[\Delta N_3]$ with $c_\pi = \mathbb{E}_\pi[\Delta\mathcal{S}]$, and for laws supported on the oriented double-star family below, $c_\pi > 2/3$.

11.3.6.1 Proof: Action-Complexity Proportionality

Derivation of the Proportionality Constant from Curvature Summation

I. Action Definition Let $I_0 = \{e \in E_0 : e \text{ meets } u \text{ or } v\}$. An old-to-old path using w can replace $u - w - v$ by $u - v$, and old paths remain available. Thus $\bar{d}_1|_{V_0 \times V_0} = \bar{d}_0$. Only the laws at u, v change among old vertices (**Lazy Causal Measure**). For $e \in E_0 \setminus I_0$, its two marginals and its ground costs on their supports are identical before and after the update, so $K_1(e) - K_0(e) = 0$. Hence

$$\begin{aligned}\Delta\mathcal{S} &= \sum_{e \in E_0} (K_1(e) - K_0(e)) + K_1(v, w) + K_1(w, u) \\ &= \sum_{e \in I_0} (K_1(e) - K_0(e)) + K_1(v, w) + K_1(w, u).\end{aligned}$$

II. Cycle and incident-edge contributions

The exact decomposition into the central edge, the two new edges and the other old incident edges is

$$\Delta\mathcal{S} = K_1(u,v) - K_0(u,v) + K_1(v,w) + K_1(w,u) + \sum_{e \in I_0 \setminus \{(u,v)\}} (K_1(e) - K_0(e)).$$

If the tree ground-metric premise holds, the central term equals $W_1^{(0)} - W_1^{(1)} > 0$ (**Cost Contraction (Phase 3)**). The fresh vertex adds exactly one directed triangle because its only incident edges are $v \to w$ and $w \to u$, and the return from u to v closes it. Thus $\Delta N_3 = 1$. A common coefficient per insertion would require control of the complete sum above.

III. Error of a cycle-only estimate

Let $C_\gamma = K_1(u,v) - K_0(u,v) + K_1(v,w) + K_1(w,u)$. The error of the cycle-only estimate satisfies

$$\left|\Delta\mathcal{S} - C_\gamma\right| \le \sum_{e \in I_0 \setminus \{(u,v)\}} |K_1(e) - K_0(e)|.$$

IV. Single-edge example

Start with $0 \to 1$. In vertex order $(0,1)$ the laws are $(2/3, 1/3)$ and $(1/3, 2/3)$. The excess $1/3$ at 0 moves to 1 at unit cost, and the difference of expectations against $\bar{d}(\cdot\,, 1)$ is $1/3$. Hence $W_1(0,1) = 1/3$ and $\mathcal{S}_0 = 1 - 1/3 = 2/3$. Adding $1 \to 2 \to 0$ produces an oriented triangle, so each law is $(1/3, 1/3, 1/3)$. All three costs vanish. Thus

$$\mathcal{S}_1 = 3(1-0) = 3, \qquad \Delta\mathcal{S} = 3 - \tfrac{2}{3} = \tfrac{7}{3}.$$

V. Chain example

Start instead with $0 \to 1 \to 2$. The old laws in vertex order $(0,1,2)$ are

$$\mu_0 = (2/3, 1/3, 0), \qquad \mu_1 = (1/3, 1/3, 1/3), \qquad \mu_2 = (0, 1/3, 2/3).$$

For both edges the excess $1/3$ at 0 moves to 2 across distance two. The dual potential $\bar{d}_0(\cdot\,, 2)$ gives $2/3$ for each difference of expectations, so both optimal costs are $2/3$ and $\mathcal{S}_0 = 2(1 - 2/3) = 2/3$. Now add $1 \to 3 \to 0$. The new laws in vertex order $(0,1,2,3)$ are

$$\mu_0 = \mu_3 = (1/3, 1/3, 0, 1/3), \qquad \mu_1 = (1/3, 1/3, 1/6, 1/6), \qquad \mu_2 = (0, 1/3, 2/3, 0).$$

Keep the common mass at each site and ship the residual as follows. The potential $f = \bar{d}_1(\cdot\,, 2)$ has values $(2, 1, 0, 2)$ and is 1-Lipschitz. Each dual bound equals the displayed feasible transport cost:

edge	residual shipments	cost	dual bound
$(0,1)$	$3 \to 2 : 1/6$	$2(1/6) = 1/3$	$2(1/6) = 1/3$
$(1,2)$	$0 \to 2 : 1/3, \quad 3 \to 2 : 1/6$	$2(1/3 + 1/6) = 1$	$2(1/3 + 1/6) = 1$
$(1,3)$	$2 \to 3 : 1/6$	$2(1/6) = 1/3$	$2(1/6) = 1/3$
$(3,0)$	none	0	0

Use $-f$ for $(1,3)$ and f for the other edges. Consequently

$$\mathcal{S}_1 = \tfrac{2}{3} + 0 + \tfrac{2}{3} + 1 = \tfrac{7}{3}, \qquad \Delta\mathcal{S} = \tfrac{7}{3} - \tfrac{2}{3} = \tfrac{5}{3}.$$

VI. Ensemble coefficient on the specified event class

For a uniform total degree bound Δ on the old endpoints, each of u, v has at most Δ incident edges. Their incident sets both contain (u, v), so $|I_0| \leq 2\Delta - 1$. Every edge curvature satisfies $-2 \leq K \leq 1$ since $0 \leq W_1 \leq 3$ and the endpoints are one hop apart (**Metric Necessity**). Thus each old difference has absolute value at most $1 - (-2) = 3$, and each new curvature has absolute value at most 2. By Step I,

$$|\Delta\mathcal{S}| \leq 3|I_0| + 2 + 2 \leq 6\Delta + 1.$$

For any probability law π supported on this class, $\mathbb{E}_\pi[|\Delta\mathcal{S}|] \leq 6\Delta + 1$. More generally, for any insertion law with an integrable action increment, set $c_\pi = \mathbb{E}_\pi[\Delta\mathcal{S}]$. Then

$$\mathbb{E}_\pi[\Delta\mathcal{S}] = c_\pi = c_\pi\, \mathbb{E}_\pi[\Delta N_3], \qquad \Delta N_3 = 1.$$

For the exact double-star host, take p incoming leaves $x_i \to u$ and q outgoing leaves $v \to y_j$ with central edge $u \to v$ and no other old edges. Add $v \to w \to u$ with w fresh. Set $a = 1/[3(p+1)]$, $b = 1/[3(q+1)]$, $I_p = \mathbf{1}_{\{p>0\}}$, and $I_q = \mathbf{1}_{\{q>0\}}$. The retained-edge curvature gain is $\min(a, b) + I_p a + I_q b$ by the double-star transport calculation (**Cost Contraction (Phase 3)**).

For $p > 0$, the leaf law is $\mu_{x_i} = 2\delta_{x_i}/3 + \delta_u/3$. After cancellation with the law at u, the residual mass at x_i is $2/3 - 1/(3p)$ before insertion and $2/3 - a$ after insertion. Every deficit is at distance two from x_i, so both optimal costs equal twice their respective residual masses. The total curvature change on the p incoming edges is

$$p\left[2\left(\tfrac{2}{3} - \tfrac{1}{3p}\right) - 2\left(\tfrac{2}{3} - a\right)\right] = 2pa - \tfrac{2}{3} = -2a.$$

The sum is zero for $p = 0$. Reversing all orientations exchanges the two equal-weight sectors, giving the outgoing contribution $-2I_q b$. The law at w is uniform on u, v, w. After cancellation in the comparisons with v and u, the leaf excesses have masses qb and pa, respectively, and all their deficit destinations are at distance two. Thus the two new curvatures are $1 - 2qb$ and $1 - 2pa$. Combining the contributions yields

$$\begin{aligned}\Delta\mathcal{S}(p, q) &= \min(a, b) + I_p a + I_q b - 2I_p a - 2I_q b + 2 - 2pa - 2qb \\ &= \boxed{\tfrac{2}{3} + \min(a, b) + (2 - I_p)a + (2 - I_q)b > \tfrac{2}{3}}.\end{aligned}$$

The formula is bounded above by $7/3$ since $a, b \leq 1/3$ and $pa = 1/3 - a$, $qb = 1/3 - b$. Every probability law supported on this family therefore has an integrable increment and strictly positive $c_\pi > 2/3$.

Q.E.D.

11.3.6.2 Commentary: Geometric Quantum

Identification of the 3-Cycle via Action-Complexity Proportionality

The action sums the edge curvatures (**Causal Ollivier-Ricci Curvature**). Each edge calculation uses the full past, center, and future weights (**Lazy Causal Measure**). Conditioning the sector choice on the center and future would instead use weights $\alpha/(\alpha+\beta)$ and $\beta/(\alpha+\beta)$, with the empty future sector returned to the center. That choice defines a separate comparison on the same undirected ground metric.

Both worked updates adjoin a fresh w and add $v \to w$ and $w \to u$ around the retained edge $u \to v$. The scheduler example instead begins with the two-path $v \to w \to u$ and adds the single return edge $u \to v$ (**Monotonicity Verification**). The fresh-vertex update changes the vertex set and two edge slots. The scheduler closure changes a single edge slot on the existing vertex set.

Each of the two worked fresh-vertex updates creates one directed triangle. The single-edge example changes the action by $7/3$, and the chain example changes it by $5/3$. Both begin with action $2/3$. The distinction

comes from the endpoint laws and the contributions of old incident edges, which enter the total action along with the two new edges.

The mean coefficient c_π describes a specified distribution of insertion events with integrable action increments. Its value can depend on the distribution of local environments. A response law for a physical ensemble would require identifying that law and controlling its mean and variation, including the effects of the old incident edges. The exact decomposition specifies the contributions that an ensemble calculation would have to retain for each update.

11.3.6.3 Calculation: Monotonicity Verification

Verification of Curvature Monotonicity via Graph Augmentation and Linear Programming

Verification of the curvature monotonicity and scaling laws established by **Action-Complexity Proportionality** is based on the following protocols:

1. **Measure Dilution Check:** The algorithm computes the lazy causal measures on the augmented graph to confirm positive shared mass across the added 3-cycle.
2. **Cost Contraction Check:** The protocol solves the optimal transport problem using linear programming to confirm a strict decrease in Wasserstein distance upon augmentation.
3. **Scaling Exponent Check:** The metric estimates the proportionality constant and scaling behavior in the sparse causal regime to validate the curvature monotonicity bounds.

```
import numpy as np
from scipy.optimize import linprog
import networkx as nx

def lazy_mu(u, G, alpha=1/3, beta=1/3):
    """
    Lazy causal measure mu_u (Measure Dilution (Phase 1) Sec.11.3.3).
    Reassigns beta if empty; dilution post-add (n^-=n_u^- +1).
    """
    N_plus = list(G.successors(u))
    N_minus = list(G.predecessors(u))
    n_plus = len(N_plus)
    n_minus = len(N_minus)
    mu = {u: alpha}
    if n_plus == 0:
        mu[u] += beta
    else:
        for w in N_plus:
            mu[w] = beta / n_plus
    if n_minus == 0:
        mu[u] += beta
    else:
        for w in N_minus:
            mu[w] = beta / n_minus
    return mu

def w1_linprog(mu_source, mu_target, dist_dict, nodes):
    """
    W_1 via linprog (Cost Contraction (Phase 3) Sec.11.3.5: Cost Contraction).
    """
    n = len(nodes)
    c = []
```

```
    inf_indices = []
    idx = 0
    # Construct cost vector
    for i, x in enumerate(nodes):
        for j, y in enumerate(nodes):
            d = dist_dict.get((x, y), np.inf)
            if np.isinf(d):
                inf_indices.append(idx)
                c.append(1e6)
            else:
                c.append(d)
            idx += 1
    c = np.array(c)

    # Equality constraints for marginals
    A_eq = np.zeros((2*n, n**2))
    b_eq = np.zeros(2*n)
    for i in range(n):
        for j in range(n):
            A_eq[i, i*n + j] = 1
        b_eq[i] = mu_source.get(nodes[i], 0)
    for k in range(n):
        for i in range(n):
            A_eq[n + k, i*n + k] = 1
        b_eq[n + k] = mu_target.get(nodes[k], 0)

    bounds = [(0, None) for _ in range(n**2)]

    # Infinite distance constraints (if any)
    if inf_indices:
        A_ub = np.zeros((len(inf_indices), n**2))
        for row, col in enumerate(inf_indices):
            A_ub[row, col] = 1
        b_ub = np.zeros(len(inf_indices))
    else:
        A_ub, b_ub = None, None

    res = linprog(c, A_eq=A_eq, b_eq=b_eq, bounds=bounds, A_ub=A_ub, b_ub=b_ub, method='highs')

    if not res.success: return np.inf
    return res.fun

def format_dict(d):
    return {k: round(v, 4) for k, v in d.items()}

# --- Simulation Setup ---
alpha = 1/3
beta = 1/3
nodes = [0,1,2]

# G0: Chain 0->1->2
# (Measure Dilution (Phase 1) Sec.11.3.3 Pre-state: Disjoint neighborhoods)
G0 = nx.DiGraph([(0,1), (1,2)])
mu0_pre = lazy_mu(0, G0)
```

```
mu1_pre = lazy_mu(1, G0)
dist = {(0,0):0, (0,1):1, (0,2):2, (1,0):1, (1,1):0, (1,2):1, (2,0):2, (2,1):1, (2,2):0}
w1_pre = w1_linprog(mu0_pre, mu1_pre, dist, nodes)
K_pre = 1 - w1_pre

# G1: Add cycle 2->0
# (Measure Dilution (Phase 1) Sec.11.3.3 Post-state: Shared mass at node 2)
G1 = G0.copy()
G1.add_edge(2, 0)
mu0_post = lazy_mu(0, G1)
mu1_post = lazy_mu(1, G1)
w1_post = w1_linprog(mu0_post, mu1_post, dist, nodes)
K_post = 1 - w1_post

# --- Verification Logic ---
# 1. Verify Shared Mass (Measure Dilution (Phase 1) Sec.11.3.3)
m_w = min(mu0_post.get(2,0), mu1_post.get(2,0))
dilution_verified = (m_w > 0)

# 2. Verify Strict Inequality (Cost Contraction (Phase 3) Sec.11.3.5)
contraction_verified = (w1_post < w1_pre - 1e-6) # explicit tolerance

# 3. Verify Sparse Scaling (Corollary 11.3.6)
m_w_sparse = beta / (0.087 + 1)  # Ch. 5 deg~=0.087 dilution
delta_k_sparse = m_w_sparse * 1.2  # Est save ~1.2 avg \bar{d}

# --- Output ---
print(f"--- State G0 (Pre-Nucleation) ---")
print(f"mu_u (0): {format_dict(mu0_pre)}")
print(f"mu_v (1): {format_dict(mu1_pre)}")
print(f"W1_pre:  {w1_pre:.4f}")
print(f"K_pre:   {K_pre:.4f}\n")

print(f"--- State G1 (Post-Nucleation) ---")
print(f"mu_u (0): {format_dict(mu0_post)}")
print(f"mu_v (1): {format_dict(mu1_post)}")
print(f"W1_post: {w1_post:.4f}")
print(f"K_post:  {K_post:.4f}\n")

print(f"--- Verification Results ---")
print(f"1. Measure Dilution (Phase 1) (Sec.11.3.3) (Shared Mass > 0):   {dilution_verified} (m_w = {m_w
print(f"2. Cost Contraction (Phase 3) (Sec.11.3.5) (W1_post < W1_pre):  {contraction_verified} (DeltaK =
print(f"3. Corollary 11.3.6 (Sparse Scaling): c ~= {delta_k_sparse:.4f} (per cycle)")
```

Simulation Results:

```
--- State G0 (Pre-Nucleation) ---
mu_u (0): {0: 0.6667, 1: 0.3333}
mu_v (1): {1: 0.3333, 2: 0.3333, 0: 0.3333}
W1_pre:  0.6667
K_pre:   0.3333

--- State G1 (Post-Nucleation) ---
mu_u (0): {0: 0.3333, 1: 0.3333, 2: 0.3333}
mu_v (1): {1: 0.3333, 2: 0.3333, 0: 0.3333}
```

```
W1_post: 0.0000
K_post:  1.0000

--- Verification Results ---
1. Measure Dilution (Phase 1) (Sec.11.3.3) (Shared Mass > 0):   True (m_w = 0.3333)
2. Cost Contraction (Phase 3) (Sec.11.3.5) (W1_post < W1_pre):  True (DeltaK = 0.6667)
3. Corollary 11.3.6 (Sparse Scaling): c ~= 0.3680 (per cycle)
```

Conclusion:

The fixture closes the existing chain $0 \to 1 \to 2$ with $2 \to 0$. The two post-update laws are equal, so the Wasserstein cost on the tracked edge drops from $2/3$ to zero and its curvature increases by $2/3$.

The separate sparse estimate uses the assigned dilution and transport factors, giving $1.2(1/3)/(1+0.087) \simeq 0.3680$. The action comparison would additionally require the contributions of the other affected edges (**Action-Complexity Proportionality**).

11.3.7 Proof: Curvature Monotonicity

Formal Verification of the Link between Topological Nucleation through Geometric Action

I. Shared mass on the new vertex

Under the hypotheses of **Curvature Monotonicity** , the shared-site masses from the dilution formula (**Measure Dilution (Phase 1)**) satisfy

$$a_u = \mu_u^{(1)}(w) = \frac{\beta}{|N_0^-(u)|+1}, \qquad a_v = \mu_v^{(1)}(w) = \frac{\beta}{|N_0^+(v)|+1}, \qquad m_w = \min(a_u, a_v) > 0.$$

II. Optimal transport cost

The coupling in **Transport Feasibility (Phase 2)** places mass m_w on (w, w) and couples the residual laws. For the full-support double-star metric, subtracting the two optimal costs gives the difference below (**Cost Contraction (Phase 3)**).

$$W_1^{(0)} - W_1^{(1)} = m_w + a_u \mathbf{1}_{N_0^-(u)\neq\emptyset} + a_v \mathbf{1}_{N_0^+(v)\neq\emptyset} \geq m_w > 0.$$

III. Curvature of the retained edge

Since $\bar{d}_0(u,v) = \bar{d}_1(u,v) = 1$, subtraction of the two curvature values yields (**Causal Ollivier-Ricci Curvature**)

$$\begin{aligned} K_1(u,v) - K_0(u,v) &= \left(1 - \frac{W_1^{(1)}}{\bar{d}_1(u,v)}\right) - \left(1 - \frac{W_1^{(0)}}{\bar{d}_0(u,v)}\right) \\ &= W_1^{(0)} - W_1^{(1)} = m_w + a_u \mathbf{1}_{N_0^-(u)\neq\emptyset} + a_v \mathbf{1}_{N_0^+(v)\neq\emptyset} > 0. \end{aligned}$$

IV. Relation to the total action

Let I_0 be the set of old edges meeting u or v. Subtracting the retained-edge term from the action decomposition gives (**Action-Complexity Proportionality**)

$$\Delta\mathcal{S} - \big(K_1(u,v) - K_0(u,v)\big) = K_1(v,w) + K_1(w,u) + \sum_{e \in I_0 \setminus \{(u,v)\}} \big(K_1(e) - K_0(e)\big).$$

Q.E.D.

11.3.Z Implications and Synthesis

Monotonicity Theorem

For the full-support double-star metric and the specified fresh-vertex insertion, the transport cost on the retained edge decreases by the shared mass and the two contributions from nonempty sectors. Its curvature increases by the same amount because the edge stays at unit distance (**Curvature Monotonicity**).

The global action sums curvature over every edge (**Discrete Einstein-Hilbert Action**). After the insertion, its change includes both new edges and any altered curvature on old incident edges. The worked examples give distinct total increments for the same cycle-count increase (**Action-Complexity Proportionality**).

The retained-edge comparison quantifies one way a local topological rewrite changes the discrete geometry. A comparison with a continuum geometric field uses the discrete curvature data and a choice of how to average them. The next chapter develops the reconstruction and tensorial averaging maps for that comparison.

11.4 Formal Synthesis

End of Chapter 11

The construction of a rigorous discrete differential geometry upon the foundation of the causal graph relies on the **GHW Metric** as the ruler of causal space. Within this metric space, the **Lazy Causal Measure** is employed to define the volume.

This volume measure defines the local geometry. Specifically, the **Causal Ollivier-Ricci Curvature** is constructed from the Wasserstein transport distance between these measures. This implies that geometry is not an abstract background, but an active manifestation of causal capacity, where flat regions represent linear transmission and curved zones indicate feedback and structural integration. In the fresh-vertex double-star case, the retained edge gains curvature (**Curvature Monotonicity**). The global action also includes the remaining incident-edge contributions. Yet, this introduces a deep physical friction: the discrete curvature is fundamentally non-local, leaving the local differential field equations of gravity as an effective approximation.

We now possess a fully defined geometric spacetime that arises directly from discrete causal relations. The stage is set for the final deductive leap: demonstrating the convergence to a continuous manifold. We turn next to **Chapter 12**, where the convergence of the discrete causal graph to a smooth, continuous space will be proved.

Table of Symbols

Symbol	Description	Context / First Used
$d_{GH}(X, Y)$	Gromov-Hausdorff distance	Sec.11.1.1.1
$d_H(A, B)$	Hausdorff distance	Sec.11.1.1.1
$W_1(\mu_X, \mu_Y)$	Wasserstein-1 transport metric	Sec.11.1.1.1
d_{GHW}	Gromov-Hausdorff-Wasserstein metric	Sec.11.1.1.1
$\bar{d}(u, v)$	Undirected shortest-path metric	Sec.11.1.2
$N^+(u), N^-(u)$	Future/Past causal neighborhoods	Sec.11.2
α	Laziness parameter (self-mass)	Sec.11.2

Symbol	Description	Context / First Used
β	Neighborhood mass parameter $((1-\alpha)/2)$	Sec.11.2
μ_u	Lazy causal probability measure for vertex u	Sec.11.2.1.1
$\mathbb{I}[\cdot]$	Indicator function	Sec.11.2.1.1
$K(u,v)$	Causal Ollivier-Ricci curvature	Sec.11.2.2
$H(\mu_u)$	Shannon entropy of measure μ_u	Sec.11.2.3
$h(\alpha)$	Allocation entropy function	Sec.11.2.3.1
d_{dir}	Directed distance function (shown insufficient)	Sec.11.2.4.1
π	Transport coupling (joint measure)	Sec.11.3.1
m_w	Zero-cost shared mass at vertex w	Sec.11.3.3
$\Delta\mathcal{S}$	Variation in total action	Sec.11.3.2
K_{baseline}	Baseline curvature in sparse graph	Sec.11.3.2.1

Document Status

Draft Version 0.3

DOI: 10.5281/zenodo.17981900

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